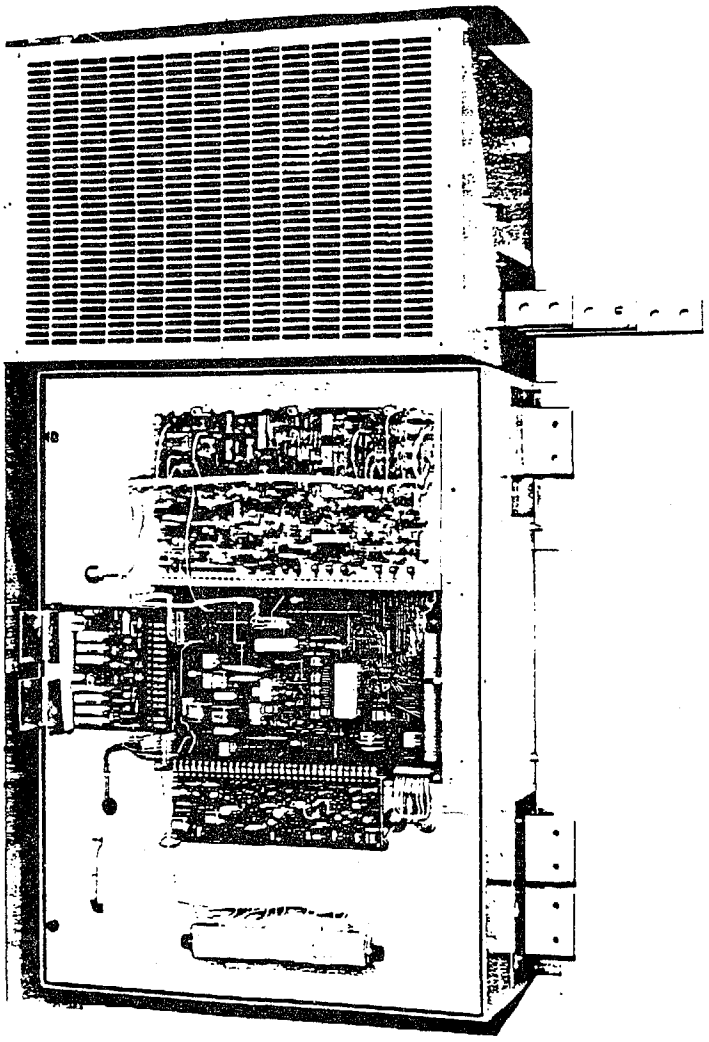
INSTRUCTIONS

ValuTrol® INDUSTRIAL DRIVE

FULL WAVE REGENERATIVE MODULE

INSTALLATION — OPERATION — MAINTENANCE



These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

GENERAL ELECTRIC

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INTRODUCTION

This Instruction Book contains helpful suggestions for placing the Valutrol drive equipment in service. It contains general information about drive operation and maintenance.

The operator and maintenance personnel should have access to a copy of this instruction book.

Additional instructions are included in the supplementary instruction publications and diagrams included in the instruction folder with the equipment.

RECEIVING, HANDLING AND STORAGE

RECEVING

The equipment should be placed under adequate cover immediately upon receipt as packing cases are not suitable for out-door or unprotected storage. Each shipment should be carefully examined upon arrival and checked with the packing list. Any shortage or damage should be reported promptly to the carrier. If required, assistance may be requested from **General Electric Company Speed Variator Products Operation, Erie, Pa.** When seeking assistance please use requisition number and model number to identify the equipment. **Telephone 814-455-3219**

HANDLING

Wall mounted power units can be transported by lift trucks with forks completely under the base using care that the unit does not tip.

STORAGE

If the equipment is not to be installed immediately, it should be stored in a clean, dry location at ambient temperatures from —20°C (—4°F) to 55°C (131°F). The surrounding air must be free of chemical and electrically conductive or corrosive contaminants.

Precautions should be taken to prevent condensation from forming within the equipment enclosure. If storage environment exceeds a 15°C (27°F) drop in temperature at 50% humidity over a 4 hour period, a space heater should be installed inside each enclosure to prevent condensation. (A 100 watt lamp can sometimes serve as a substitute source of heat). Higher humidities with smaller temperature changes will also cause condensation.

Condensation occurs when air containing some moisture is cooled below its dew point. The dew point represents saturation of the air, and is the temperature at which the moisture starts to condense into water. It is not a fixed temperature but rather is related to the initial temperature of the air and its relative humidity at that temperature. The amount of moisture that can be held in the air is related to the air temperature. The following examples illustrate some of these relationships.

TABLE I
Relationship Between Air Temperature Relative Humidity and Dew Point

AIR TEMP °F	°C	RELATIVE HUMIDITY %	WGT. OF MOISTURE IN 1 LB OF DRY AIR. GRAINS	DEW POINT °F	°C
104	40	100	345	104	40
104	40	80	270	97	36
104	40	40	130	75	24
104	40	10	32	37	3
50	10	100	54	50	10
50	10	80	42	43	6
50	10	40	21	25	4

In industrial drives, condensation is a possibility in applications where air temperature changes are large and rapid and/or the air is moist. For example, an outdoor crane operating in sunshine on a winter day, which then is shut down and parked in the shade will experience a rapid drop in temperature. This can result in condensation inside the equipment. Adding heat to keep the air temperature above its dew point can prevent condensation.

If storage temperatures below —20°C (—4°F) are likely to be present then auxiliary heat should be added in each enclosure to maintain temperature at or above —20°C. For assistance in heater size selection contact General Electric Company.

When a drive that has been in operation is shut down for either a short or extended period of time, it is recommended the environmental conditions be maintained the same as when in operation. Power, ventilation or heating and air conditioning (if used) should be left on during the downtime to prevent large changes in temperature and possible moisture condensation.

SAFETY FOR PERSONNEL AND EQUIPMENT

The following paragraphs list some general safety reminders and safety recommendations to be followed when operating or installing this equipment.

WARNING

DENOTES OPERATING PROCEDURES AND PRACTICES THAT MAY RESULT IN PERSONAL INJURY OR LOSS OF LIFE IF NOT CORRECTLY FOLLOWED.

COLOR — BLACK OR WHITE LETTERING ON RED FIELD.

CAUTION

DENOTES OPERATING PROCEDURES AND PRACTICES THAT, IF NOT STRICTLY OBSERVED MAY RESULT IN DAMAGE TO, OR DESTRUCTION OF, THE EQUIPMENT.

COLOR — BLACK LETTERING ON AMBER FIELD.

NOTE

DENOTES AN OPERATING PROCEDURE OR CONDITION WHICH SHOULD BE HIGHLIGHTED.

COLOR — BLACK LETTERING ON A WHITE FIELD.

WARNING

IMPROPER LIFTING PRACTICES CAN CAUSE SERIOUS OR FATAL INJURY.
LIFT ONLY WITH ADEQUATE EQUIPMENT AND TRAINED PERSONNEL.

WARNING: HIGH VOLTAGE

ELECTRIC SHOCK CAN CAUSE PERSONAL INJURY OR LOSS OF LIFE. CIRCUIT BREAKERS, IF SUPPLIED AS PART OF THE TOTAL SYSTEM, MAY NOT DISCONNECT ALL POWER TO THE EQUIPMENT. SEE SYSTEM ELEMENTARY DIAGRAMS. WHETHER THE AC VOLTAGE SUPPLY IS GROUNDED OR NOT, HIGH VOLTAGE TO GROUND WILL BE PRESENT AT MANY POINTS. WHEN INSTRUMENTS SUCH AS OSCILLOSCOPES ARE USED TO WORK ON LIVE EQUIPMENT, GREAT CAUTION MUST BE USED. WHEN ONE OF THE INSTRUMENT LEADS IS CONNECTED TO THE CASE OR OTHER METAL PARTS OF THE INSTRUMENT, THIS LEAD SHOULD NOT BE CONNECTED TO AN UNGROUNDED PART OF THE SYSTEM UNLESS THE INSTRUMENT IS ISOLATED FROM GROUND AND ITS METAL PARTS TREATED AS LIVE EQUIPMENT. USE OF AN INSTRUMENT HAVING BOTH LEADS ISOLATED FROM THE CASE PERMITS GROUNDING OF THE CASE EVEN WHEN MEASUREMENTS MUST BE MADE BETWEEN TWO LIVE PARTS.

CAUTION

DO NOT REMOVE PRINTED CIRCUIT CARDS FROM THE EQUIPMENT WHILE POWER IS APPLIED. THIS CAN DAMAGE THE EQUIPMENT.

NOTE

ALWAYS READ THE COMPLETE INSTRUCTION PRIOR TO APPLYING POWER OR TROUBLE-SHOOTING THE EQUIPMENT. FOLLOW THE START—UP PROCEDURE STEP BY STEP.
READ AND HEED ALL WARNING, CAUTION AND NOTE LABELS POSTED ON THE EQUIPMENT.

CAUTION

DO NOT REMOVE INPUT POWER FROM THE DRIVE UNTIL IT HAS FULLY EXECUTED A STOP SEQUENCE, AS THIS CAN DAMAGE THE DRIVE SYSTEM.

INSTALLATION

LOCATION

DC-SCR drive power units are suitable for most factory areas where other industrial equipment is installed. They should be installed in well ventilated areas with ambient temperatures ranging from 0°C (32°F) to 40°C (104°F) and relative humidities up to 90 percent. It should be recognized; however, that since the life expectancy of any electronic component decreases with increased ambient temperature, reduction of the ambient temperature will bring about extended component life. For example, longer component life should be expected if the ambient temperature is held between 20° C (68° F) and 30° C (87° F).

Proper performance and normal operational life can be expected by maintaining a proper environment for the drive system.

Environments which include excessive amounts of one or more of the following characteristics should be considered hostile to drive performance and life:

1. Dirt, dust and foreign matter.
2. Vibration and shock.
3. Moisture and vapors
4. Temperature excursions.
5. Caustic fumes.
6. Power line fluctuations.
7. Electromagnetic interference (noise).

Totally enclosed power units should be positioned to permit heat radiation from all surfaces except the bottom; otherwise, the enclosure can be positioned as follows:

Enclosures may be placed side by side with another enclosure. Clearance at least equal to the width of the enclosure should be available in front so that the door may be fully opened for easy access.

WARNING

EXPLOSIONS OR FIRES MIGHT RESULT FROM MOUNTING DRIVE POWER UNITS IN HAZARDOUS AREAS SUCH AS LOCATIONS WHERE INFLAMMABLE OR COMBUSTIBLE VAPORS OR DUSTS ARE PRESENT. DRIVE POWER UNITS SHOULD BE INSTALLED AWAY FROM HAZARDOUS AREAS, EVEN IF USED WITH DC MOTORS SUITABLE. FOR USE IN SUCH LOCATIONS.

MOUNTING

Enclosures may be mounted on any firm, reasonably flat, vertical surface. Four holes (one in each rear corner) are provided for mounting the power module.

CAUTION

THE ENCLOSURE MUST BE MOUNTED IN SUCH A MANNER AS NOT TO RESTRICT THE AIR FLOW UP THRU THE BOTTOM OF THE ENCLOSURE, OR OUT THRU THE TOP OF THE ENCLOSURE.

CONNECTIONS

All internal electrical connections between components in DC-SCR drive power units are made at the factory of General Electric Company.

Be sure to protect the interior panel mounted components and sub-assemblies from metal particles when cutting or drilling entrances for interconnecting wiring and cables.

If additional relays, contactors, or electrical solenoids are added in the proximity of the SCR equipment enclosure, RC suppression networks should be added across the coils. A series combination of a 220 ohm resistor and a 0.5mfd capacitor in parallel with the relay coils is recommended.

NOTE

SOME SYSTEM TRANSFORMERS AND OTHER APPARATUS ARE SHIPPED SEPARATELY AND MUST BE MOUNTED AND CONNECTED TO THE SYSTEM.

WARNING

ALL MOTOR BASES AND EQUIPMENT ENCLOSURE HOUSINGS SHOULD BE CONNECTED TO THE FACTORY OR FACILITY EARTH GROUNDING SYSTEM.

NOTE

IT IS RECOMMENDED THAT THE DRIVE SYSTEM COMMON CIRCUIT BE GROUNDED AT ONLY ONE POINT. THIS MEANS THAT IF THE DRIVE REFERENCE IS SUPPLIED BY A NUMERICAL CONTROL OR PROCESS INSTRUMENT WITH GROUNDED COMMON. THE DRIVE COMMON SHOULD NOT BE GROUNDED. IF A SEPARATE POWER TRANSFORMER IS USED AND IF THE SECONDARY OF THE TRANSFORMER MUST BE GROUNDED, IT IS RECOMMENDED THAT HIGH RESISTANCE GROUNDING BE USED.

CAUTION

INSTALLATION WIRING MUST BE IN ACCORDANCE WITH THE NATIONAL ELECTRICAL CODE AND BE CONSISTENT WITH ALL LOCAL CODES. SECONDARIES OF 115 VOLT CONTROL TRANSFORMERS TYPICALLY HAVE ONE SIDE FUSED AND THE OTHER GROUNDED OR AVAILABLE FOR GROUNDING BY THE USER.

CAUTION

MEGGERING CAN DAMAGE ELECTRONIC COMPONENTS. DO NOT MEGGER OR HI-POT WITHOUT CONSULTING THE SPEED VARIATOR PRODUCTS OPERATION, GENERAL ELECTRIC CO.

CAUTION

DO NOT CONNECT ANY EXTERNAL CIRCUITS OTHER THAN SHOWN ON THE ELEMENTARY DIAGRAM, SUCH AS AMMETERS ON THE SHUNT OR VOLTMETERS ON THE TACHOMETER BECAUSE THE PERFORMANCE OF THE DRIVE SYSTEM MAY BE DEGRADED.

CAUTION

DO NOT USE POWER FACTOR CORRECTION CAPACITORS WITH THIS EQUIPMENT WITHOUT CONSULTING THE SPEED VARIATOR PRODUCTS OPERATION, GENERAL ELECTRIC CO. DAMAGE MAY RESULT FROM HIGH VOLTAGES GENERATED WHEN CAPACITORS ARE SWITCHED.

Before power is applied to the drive system, checks should be made to see that all internal connections are tight, that plug in printed circuit cards in the optional regulator rack are fully seated and that all open relays and contactors operate freely by hand. Check that the equipment is clean and no metal chips are present.

MAINTENANCE

Periodically inspect and maintain the equipment protective devices (particularly air filters when supplied) per instructions in this section. Check all electrical connections for tightness; look for signs of poor connections and overheating (arcing or discoloration).

FANS AND FILTERS

On force ventilated drives, the power unit contains a fan and perhaps an air filter in the intake of the enclosure and/or equipment inside the enclosure.

Inspect the fan at regular intervals to see that it is operating properly. Check for excessive noise and vibration, loose fan blades and for overheating of the motors. Keep the fan blades clean.

If the fan does not operate, replace the fan and integral motor with a unit with the same catalog number.

Clean and/or replace air filter as appropriate depending on the accumulation of dirt for the type supplied.

To clean metal filters, flush only with warm water, dry and recoat lightly with RP® Super Filter Coat or equivalent (light oil) or replace the filter.

Be sure to install filters with air flow direction as indicated on the filter.

DC MOTORS

Maintenance instructions covering brushes, commutator and lubrication are in GEK-2304 which is found elsewhere in the instruction book.

PRINTED CIRCUIT CARDS

Printed circuit cards normally do not require maintenance except to keep them clean and tightly secured to their respective terminal boards or tightly plugged in the optional modification rack receptacles. Clean as follows:

1. Dry Dust — Vacuum clean, then blow with dry filtered compressed air (low pressure supply).

2. Oily Dirt — Certain components (electrolytic capacitors, switches, meters, potentiometers and transformers) can be damaged by solvent, so its use is not recommended. If absolutely necessary, use solvent sparingly on a small brush and avoid above components. Clean contact terminals with dry non-linting cloth after solvent has been used. Recommended solvents: Freon* RE or TF.

3. If the card is badly contaminated or corroded, replace.

SILICON CONTROLLED RECTIFIERS

Keep SCR's and heatsink free from dirt, oil or grease, since any accumulation of dirt may cause overheating. Clean as follows:

1. Dry Dust — Vacuum clean, then blow with dry, filtered compressed air (low pressure).

CAUTION

SOLVENT CAN HARM NON-METAL COMPONENTS.

2. Oil Dirt — Use dry or barely moist (with solvent) non-linting cloth. Repeat until cloth remains clean. All contact tips must be cleaned with dry non-linting cloth after solvent has been used. Recommended solvents: Freon RE or TF.

CONTROL DEVICES

Inspect all relays and contactors at regular intervals and keep them free from dirt, oil or grease. Check for freedom of moving parts, corrosion, loose connections, worn or broken parts, charred insulation or odor, proper contact pressure and remaining wear allowance on contacts. **Do not** lubricate the contacts as lubrication shortens their life.

Both copper and silver contacts will become darkened and somewhat roughened in normal operation. This does not interfere with their performance, and does not indicate that the contacts should be filed. In general, contacts will not need attention during their normal life, but if prominent beads form on the surfaces due to severe arcing, the contact faces may be dressed with a fine file. **Do not** use sand paper or emery cloth.

Any contact that is worn to the point where contact wipe or pressure is lost should be replaced. Contactor shunts which are badly frayed or broken should also be replaced.

Cleaning procedure is the same as previously given for SCR and heatsink.

*Trademark of E.I. DuPont Co.

INSTRUCTION INFORMATION

This instruction folder furnished with the equipment includes detailed instructions and diagrams applicable for each specific drive system.

In addition to this general instruction the folder includes instruction for the motor(s) and other components furnished. Start-up and troubleshooting guides are included. All instructions and the accompanying diagrams should be consulted before applying power to the system.

THE FOLLOWING INFORMATION IS OF PARTICULAR IMPORTANCE

TYPES OF DIAGRAMS

Different types of control diagrams are provided for specific purposes. The type of control diagram is noted in the title block of each diagram sheet.

The major types of diagrams are **Elementary**, (sometimes called schematic), **Layout** or **Connection** and **Interconnection.**

The **Elementary** diagrams represent (in symbolic form) the fundamental operation and relationship of the electrical parts of a system. These diagrams are drawn in such a manner that the operation of the control system is easily understood. Mechanical relationships of control devices are subordinated to simple presentation of the electrical circuits. Connections made between control devices and power devices within the enclosure are also shown in this type diagram.

The elementary diagram also identifies adjustments, signals and test points. In this instruction book, adjustments are CAPITALIZED. Example: **FMAX** (Maximum motor field adjustment).
Signals and test points are CAPITALIZED. Example: CFB (Current Feedback)

The **Layout** or **Connection** diagram, when supplied is one which shows relative physical position of the devices as well as other electrical components located within the same enclosure.

In many cases the **Elementary** diagram will be combined with the **Interconnection** diagram. For a complex system, separate **Interconnection** diagram will be furnished, which will show the type and number of connections to be made between major components of the system such as the power unit, motor, operator's station, the plant power source, auxiliary devices and other electrical machines. In some cases the **Interconnection** information may be presented in tabular form.

GENERAL DESCRIPTION

Input 3 phase power is fed through the incoming bus at the top of the enclosure, through current transformers and enters the power conversion module where it is converted to DC adjustable voltage. DC current is fed through a shunt and a DC contactor to the DC motor armature.

The basic elements of the full wave, regenerative, SCR Drive are shown in Figure 1.

The speed of the motor is proportional to the DC voltage applied to its armature. Speed may be measured by a tachometer generator directly connected to a DC motor.

The remainder of the control is manufactured on eight (8) removable printed circuit boards. These are the Power Supply (PSC), the Main Control card (MCC), the Auxiliary Control card (ACC), the Interface card (IFC), the Motor Field card (MFC), or Motor Field Exciter (MFE), the Diagnostic card (DGC), the Auxiliary Interconnect card (AIC) and the Regulator Buffer card (RBC).

Signal level power for the control is taken from the three phase input through control fuses to the control power transformer (CPT). This transformer is fitted with two isolated secondary windings: (1) 115V AC to operate the coils of the M, PR, C on and external reset relays; (2) The second winding is a 50 volt AC tapped secondary which provides the AC input to the power supply card.

POWER SUPPLY CARD (PSC)

The power supply card rectifies the AC input and provides regulated plus and minus 20 volts for the printed circuit cards. Unregulated plus and minus 30 volts DC is also provided to drive the static logic switches and the "control on" function. All of the DC outputs are fused to protect the power supply card against overloads. The regulated plus and minus 20V DC outputs are protected against overvoltage conditions caused by a power supply card failure.

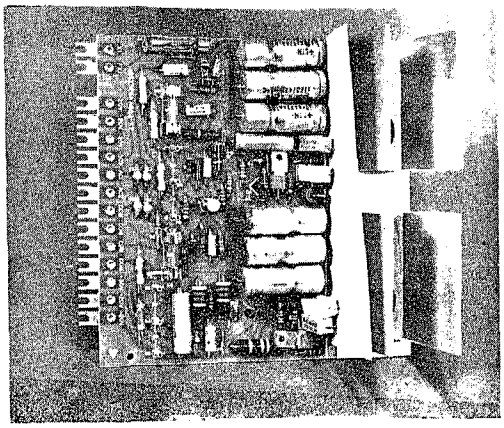


FIGURE 3
POWER SUPPLY CARD

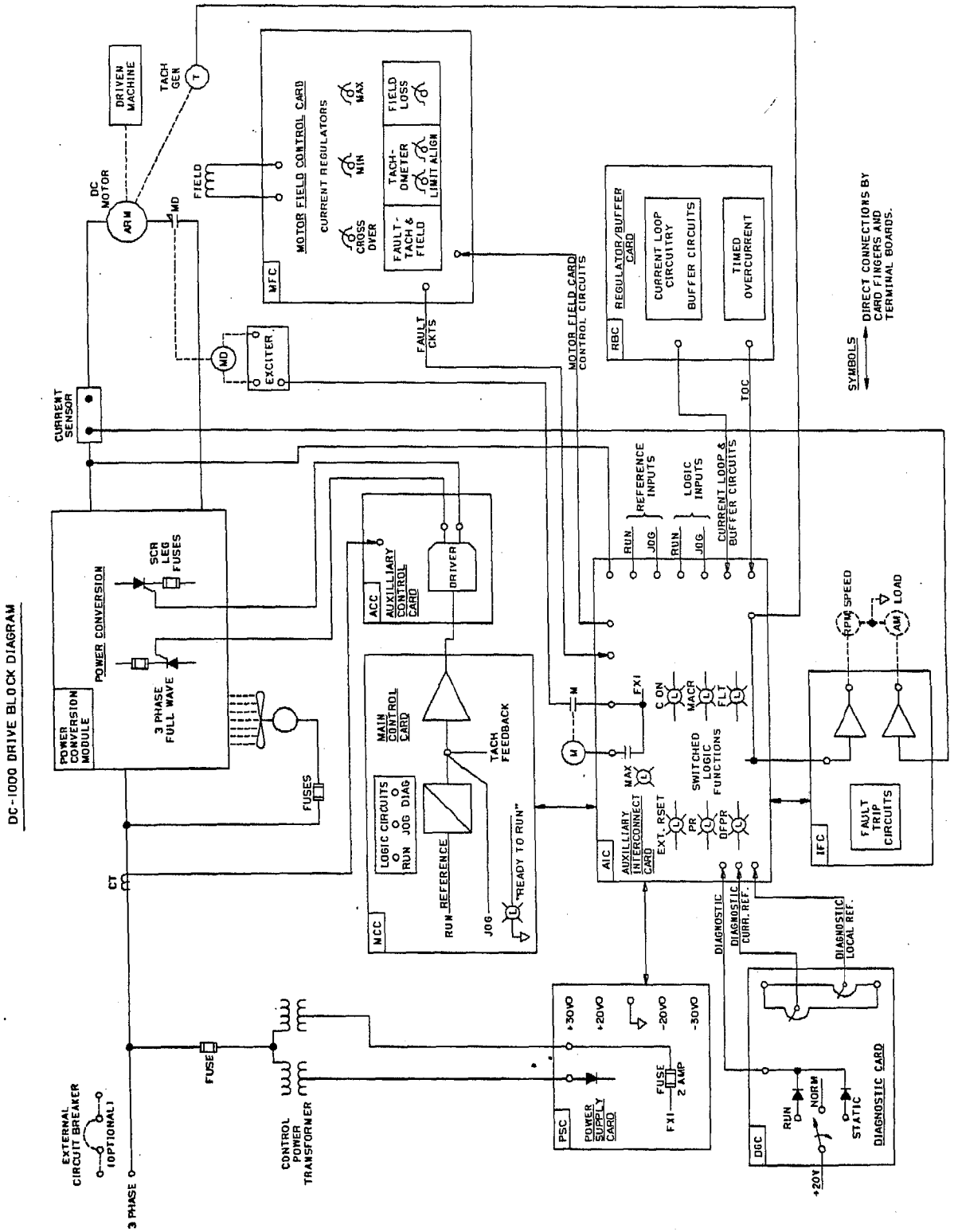


FIGURE 1
DRIVE BLOCK DIAGRAM (36C766718AA)

1. PLENUM WITH FAN
2. FAN MOTOR FUSES (BEHIND SCREEN)
3. MAIN CONTROL CARD (MCC)
4. AUXILIARY INTERCONNECT CARD (AIC)
5. 2TB
6. REGULATOR/BUFFER CARD (RBC)
7. INTERFACE CARD (IFC)
8. DIAGNOSTIC CARD (DCG)
9. POWER SUPPLY CARD (PSC)

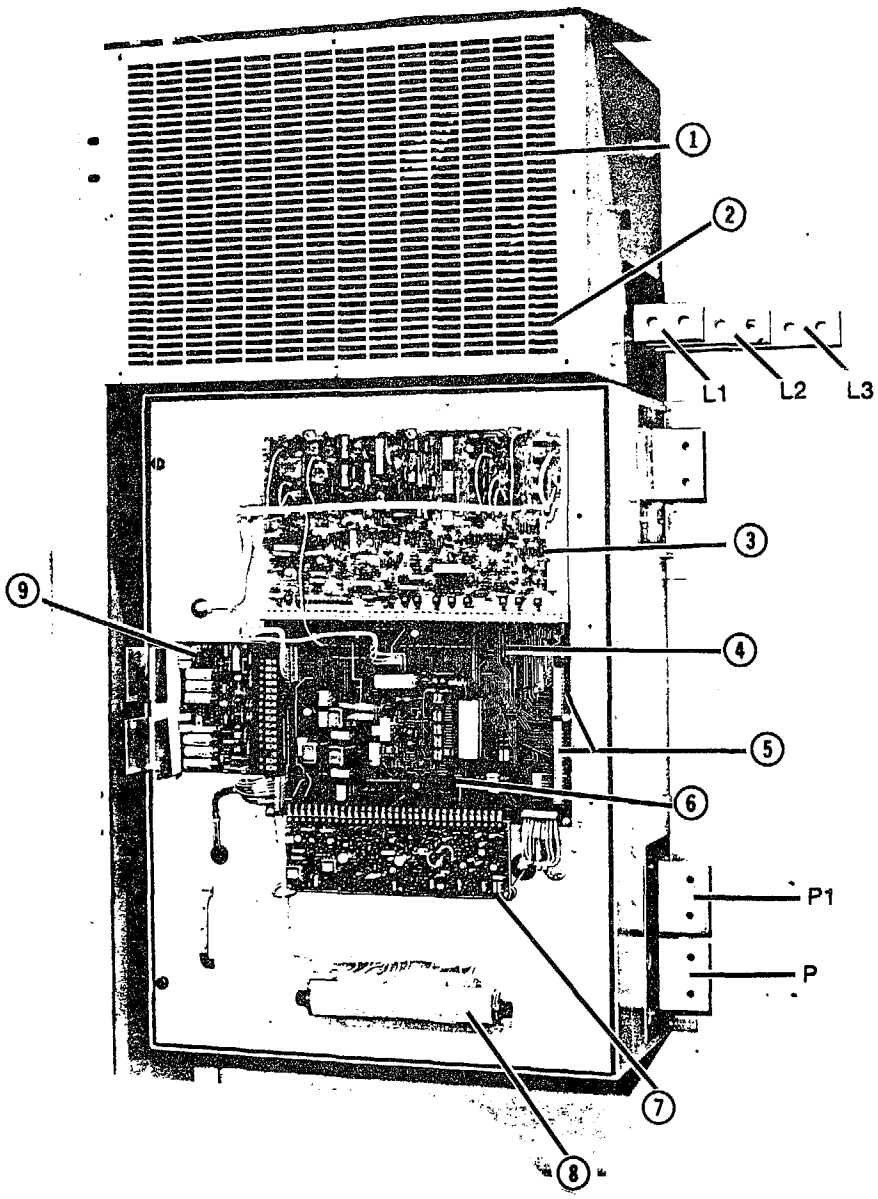


FIGURE 2
POWER UNIT

1. DC SNUBBER ASSEMBLY
2. AC SNUBBER ASSEMBLY
3. AUXILIARY CONTROL CARD (ACC)
4. 9TB
5. M RELAY
6. CONTROL POWER TRANSFORMER (CPT)
7. TBA (CURRENT TRANSFORMER TERMINAL BOARD)

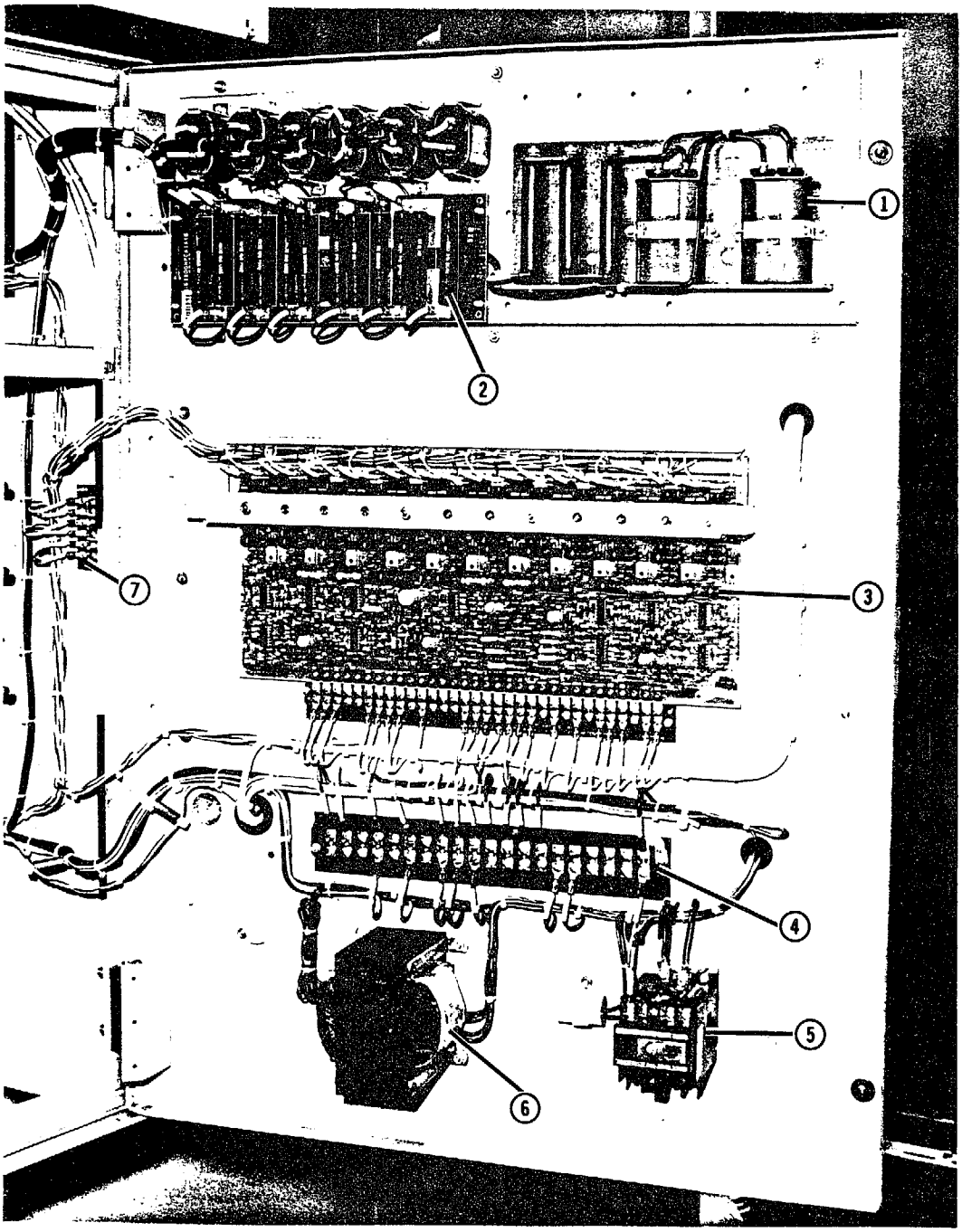


FIGURE 2A
POWER UNIT (door-inside view)

1. AC BUS
2. N (NEGATIVE DC BUS)
3. P1 (POSITIVE DC BUS)
4. P (POSITIVE DC BUS — SHUNT OMITTED)
5. BUS STRAPS (MAY BE REPLACED WITH FUSES)
6. MODULE THERMAL SWITCHES
7. SC STACK ASSEMBLIES (TWO SCR's PER STACK)

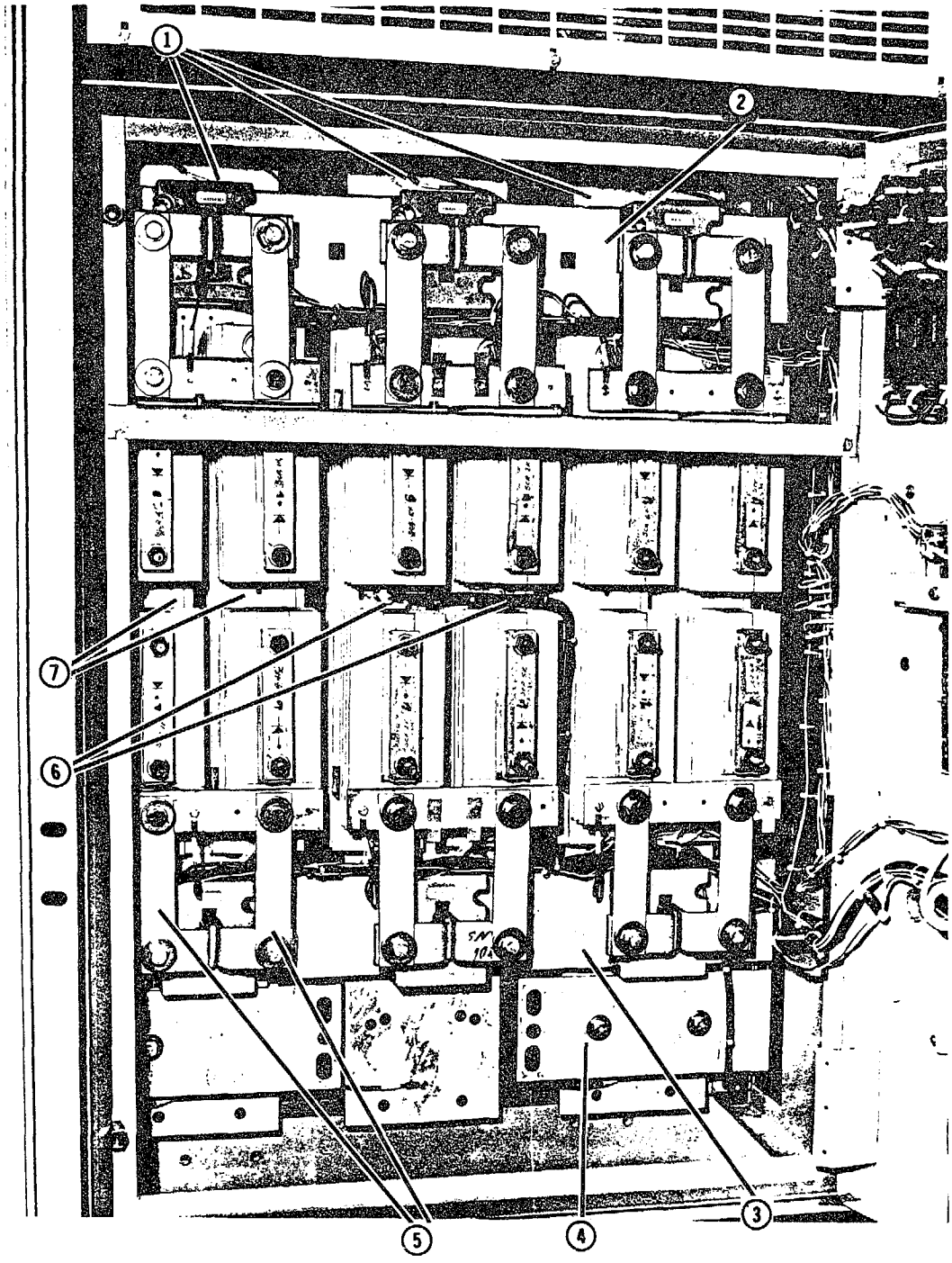


FIGURE 3B
POWER UNIT (inside view)

MAIN CONTROL CARD (MCC)

The primary purpose of the main control card is to drive the conversion module as commanded by the speed reference and feedback signals. Connections to the SCR gates is by way of the auxiliary control card (ACC).

This card also performs several additional functions such as linear timing of the reference; current limit; "Ready to Run" indicator, and various scaling and trimming adjustments.

A total of twelve (12) potentiometers are provided on this card, ten (10) of which are accessible from the front of the controller, The eleventh potentiometer, the card zero adjustment **ZERO ADJ**, is preset at the factory and should normally not be disturbed. The twelfth potentiometer is the line impedance compensation adjustment, **LINE**. The ten accessible potentiometers are:

DAMP	**MAX SPEED**	**MIN SPEED**
CUR LIMIT	**GAIN**	**REF SCALE**
CEMF LIMIT	**RESPONSE**	**LIN TIME**
COMP		

When the drive is first placed into operation the actual top speed may be different from what might be expected due to minor variations between tachometers. By adjusting the **MAX SPEED** potentiometer, any variations between tachometers can be eliminated without disturbing any other adjustments in the drive.

AUXILIARY CONTROL CARD (ACC)

The primary function of this card is to combine phase control signals, leg current signals, and oscillator signals and amplify the resulting pulse trains and direct them to the appropriate SCR gates by way of twelve (12) pulse transformers. One secondary function is to scale the output of three (3) current transformers (CT's) in the three phase AC power inputs to the conversion module.

Another function is to detect circulating overcurrent faults which would not be detected by the shunt in the motor armature loop.

An additional function is to generate a trip signal which suppresses SCR firing, executes a system fault, and may be used to operate an optional shunt trip circuit breaker.

INTERFACE CARD (IFC)

The primary purposes of the interface card are:

1. To provide low level isolated signals corresponding to the three phase AC line voltage, DC armature current voltage, armature current and tachometer feedback.

2. To control the start, stop and synchronizing of the drive while monitoring the system for abnormal operating conditions.

3. To provide one milliampere signals for external speed and current indication.

Other outputs provide:

1. A N.O. contact indicating a fault condition (FLT).

For those drives employing an AC tachometer, an output whose frequency is proportional to RPM is generated which may be used to drive a digital counter. A type AN AC tachometer will generate 18 pulses for each revolution.

There are (4) potentiometers on this card.

1. The **IZERO** is a bias adjustment for the current feedback output and is factory set. This control should not be disturbed.

2. **R STOP** is the drop out level of the regenerative stop sequencing circuit and is also factory set.

3. **IMET** is the calibration adjustment for the current indicator.

4. **SMET** is the calibration adjustment for the speed indicator.

Ajustments 3 and 4 will be factory set if the indicators are ordered with the drive and mounted in the power enclosure.

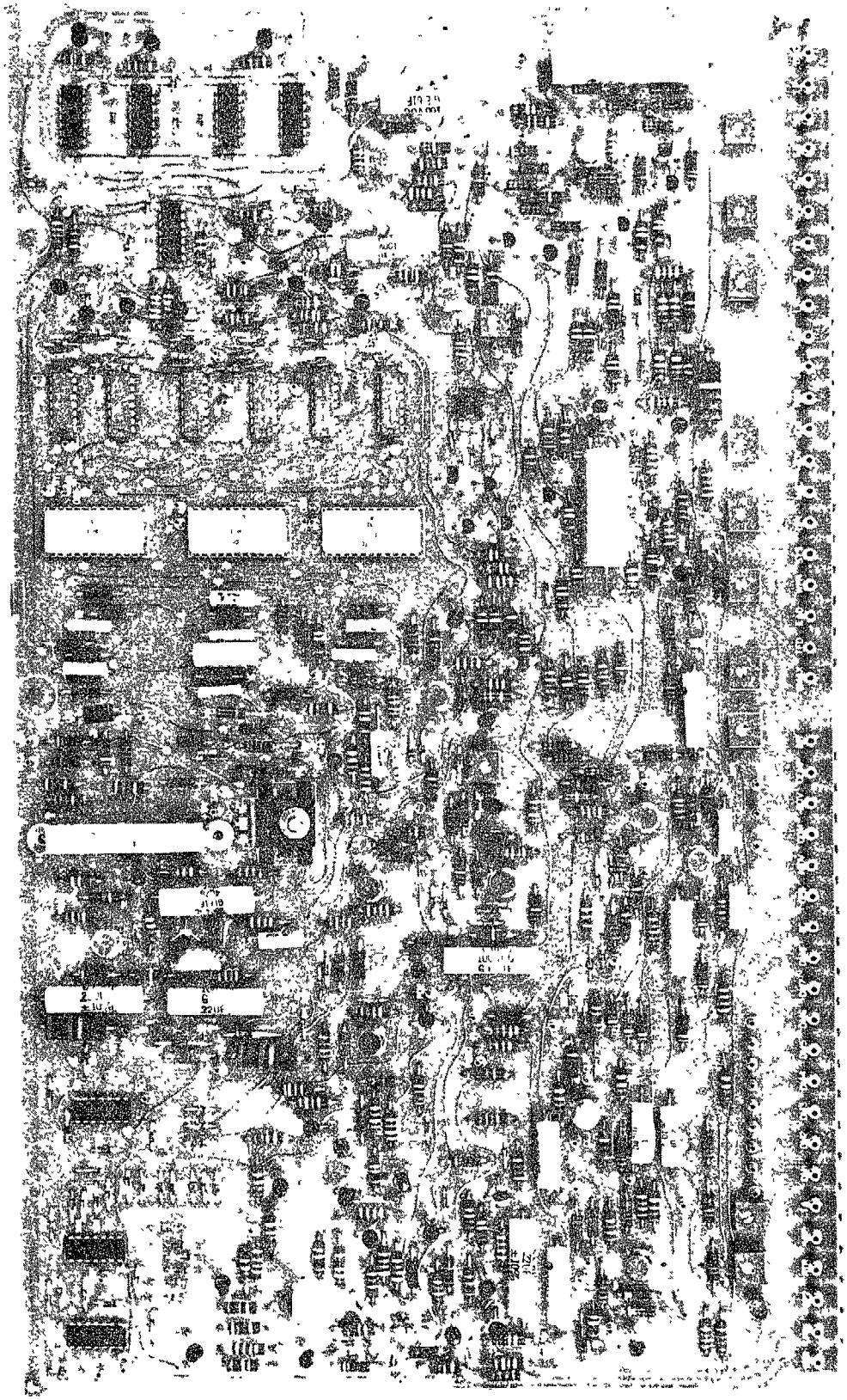


FIGURE 4
MAIN CONTROL CARD

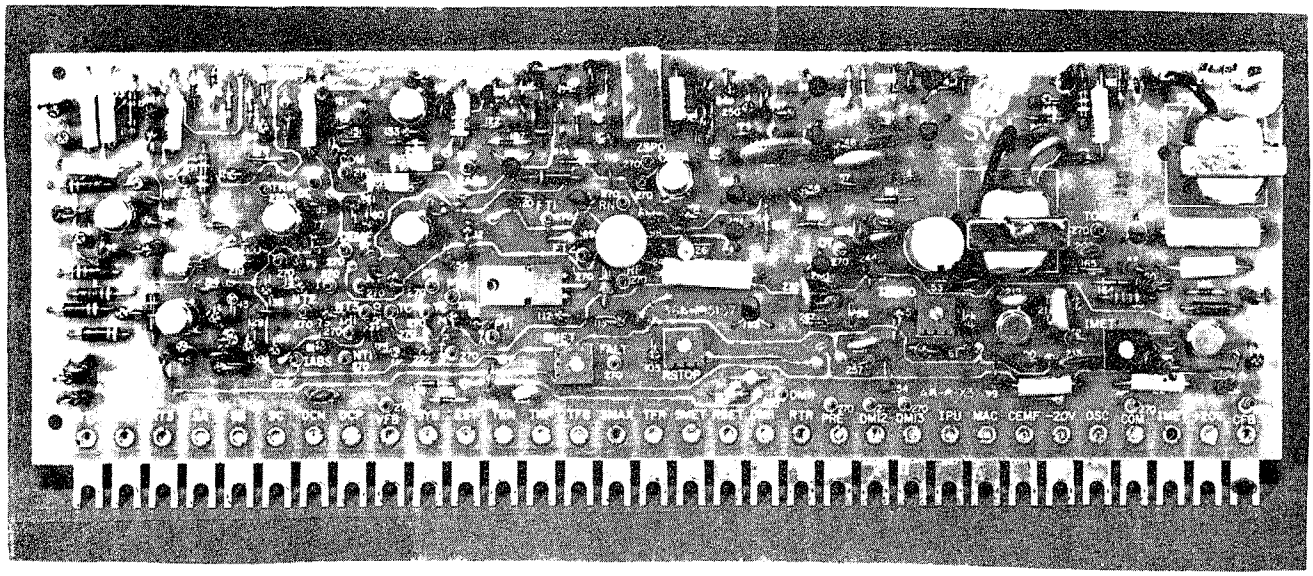


FIGURE 5
INTERFACE CARD

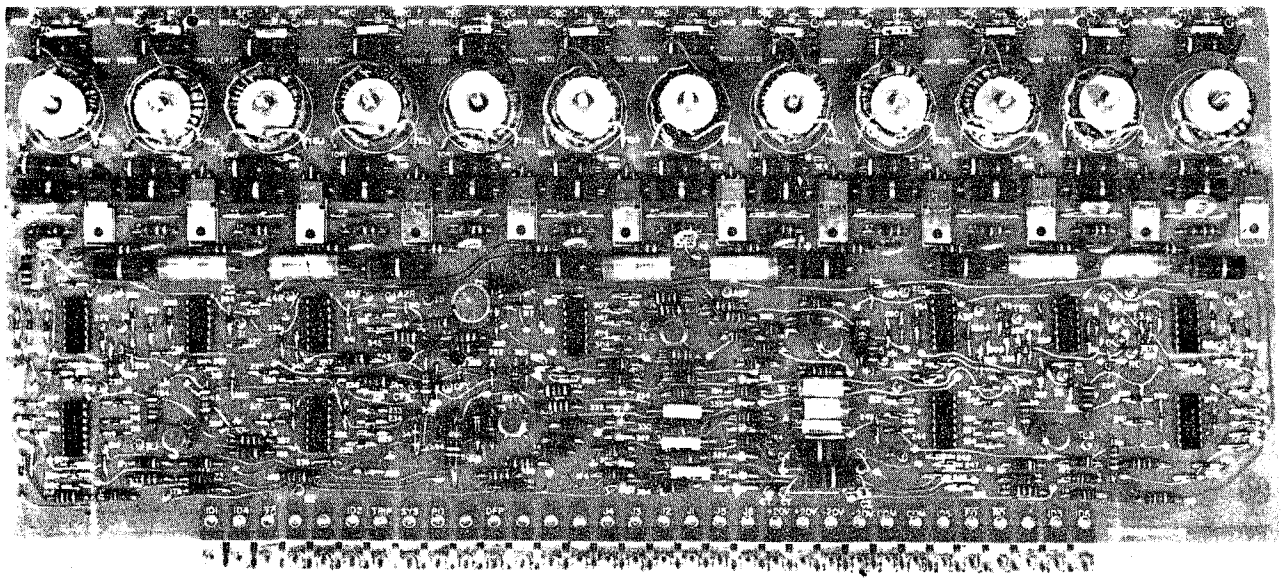


FIGURE 6
AUXILIARY CONTROL CARD

DIAGNOSTIC CARD (DGC)

The diagnostic card performs no function under normal operating conditions but will program the drive into a diagnostic run mode and diagnostic static mode for ease in initial start up and troubleshooting.

AUXILIARY INTERCONNECT CARD (AIC)

The primary purposes of the Auxiliary Interconnect card are:

1. To interconnect the Main Control card, Interface card, Power Supply card and Regulator/Buffer card into one integral assembly on the door of the enclosure.

2. To provide interfacing between the cards listed above and the Diagnostic and Auxiliary Control cards.

3. To provide I/O capability to the drive through terminal boards and/or ribbon cable connectors.

4. To provide switched logic functions with LED indication to include:

   A. Preconditioning (PR)
   B. Delayed Firing Power (DFPR)
   C. External Reset (EXT. RSET)
   D. Control ON (C ON)
   E. Fault (FLT)
   F. MAX (MAX)
   G. MACR (MACR)

REGULATOR/BUFFER CARD (RBC)

The primary purposes of the Regulator/Buffer card are:

1. To provide inner current loop electronics for stability requirements in certain applications.

2. To provide buffer circuitry for various I/O signals.

3. To provide timed overcurrent circuitry to protect the motor from extended overloads.

CONTROL FUSES

The signal power for the control is taken from the three phase input through control fuses to the control voltage transformer (not shown on the block diagram). The control fuses are used to protect the control transformer and the drive will not operate if any one of these fuses is open.

POWER CONNECTIONS

The AC power connections are three phase input at L1, L2 and L3. DC power output at P and N.

SIGNAL CONNECTIONS

All signal connections are made on the 2TB, 7TB and 6TB terminal boards.The signals appearing on 2TB and their functions are described in Table III. Refer to the system elementary diagram for details.

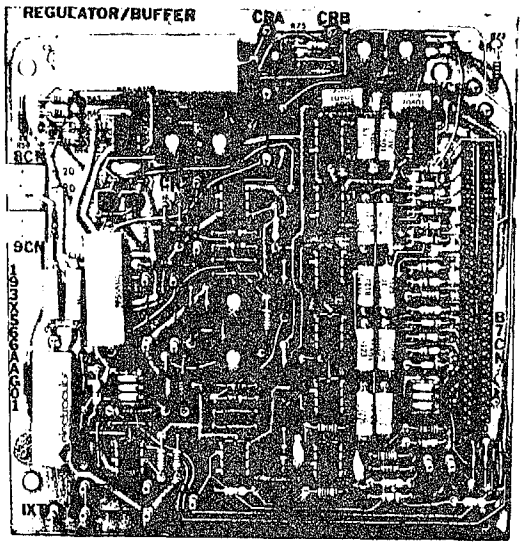


FIGURE 7
REGULATOR/BUFFER CARD

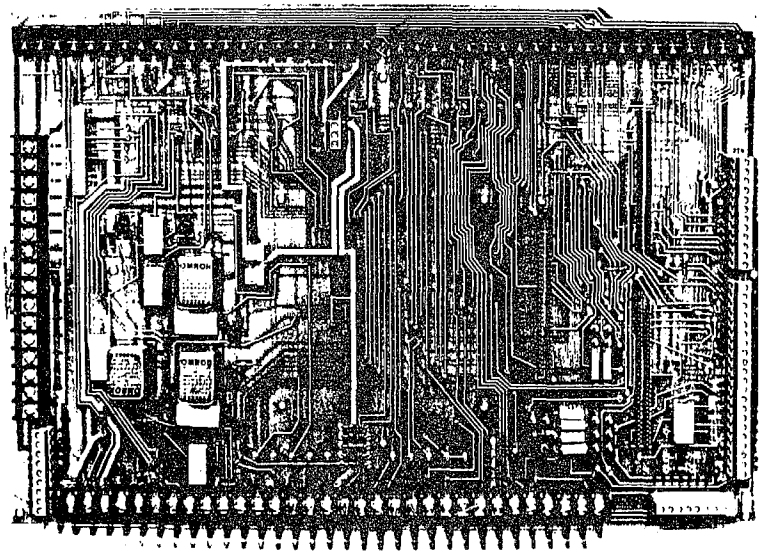


FIGURE 9
AUXILIARY INTERCONNECT CARD

TABLE III SIGNAL CONNECTIONS

2TB	NOMENCLATURE	DESCRIPTION
1	**—30V**	Unregulated negative DC voltage used as the return line for static switches RUN and JOG and possible modifications.
2,3,4	**±20V**	Regulated power supply outputs.
5,6,7	**COM**	Signal common. All signals are measured with respect to common, unless otherwise noted.
8	**EST**	External fault stop input. If EST is momentarily disconnected from common, the MD contactor will open and the motor will coast. The drive may not be restarted until the RSET line is momentarily connected to common with the pushbutton RSET on the AIC card or through use of the external RSET relay. (See ext. RSET below).
9	**SMIN**	Output from the MIN SPEED potentiometer on the Main Control card.
10	**SR**	Speed Reference Input.
11	**JOGR**	Jog Reference Input
12	**SMET**	Output to a 1ma speed instrument. The instrument is calibrated with the **SMET** potentiometer on the Interface card.
13	**IMET**	Output to a 1ma load instrument. The instrument is calibrated with the **IMET** potentiometer on the Interface card.
14,15 16,17,18 19,20 21	**SFB, CEMF** **SYSX, FC, TA** **FDR, WFR** **CRM**	Controls signals for Motor Field card.
22 23	**JOG** **RUN**	The drive will not start unless RUN or JOG are connected to —30V, either at 2TB or by special purpose logic wired into collectors 3CN and 4CN on on the AIC card. When RUN and JOG are released from —30V the drive will decelerate to a stop and open the MD contactor.
24	**CON**	The drive will not start unless the Control ON Relay is picked-up by applying +20V to C ON. If +20V is released from C ON. The MD contactor will open and the drive will coast.

TABLE III SIGNAL CONNECTIONS
(continued)

2TB	NOMENCLATURE	DESCRIPTION
25	**MACR**	For special applications. Relay MACR will be picked-up by applying +20V to 2TB-25 to disconnect the MIAC signal from the Diagnostic card to the Main Control card and Interface cards.
26	**PR**	Preconditioning relay PR must be picked-up by applying 115V AC to PR in order for the drive preconditioning to be released. Control is normally accomplished through Auxiliary Contacts on the MD contactor.
27	**DFPR**	Delayed firing power relay (DPFR) must be picked-up by applying 115V AC to DFPR in order to send firing signals to SCR's. Control is normally accomplished through contacts on N1 relay.
28,29	**FLT 8, 4**	A normally open, held closed relay contact. Under normal conditions this contact is closed. If a fault condition is detected, this contact opens.
30	**FX1**	Fused 115V AC.
31	**EXT. RSET**	115V AC control input for an external fault RSET pushbutton. By momentarily connecting 115V AC to 2TB-31, the RSET line will be taken to common by ext. RSET relay.
32	**MAX 8**	Fused 115V AC is switched to MAX 8 when relay MAX picks up.
33	**RN**	Unregulated —30V is switched to RN when relay C ON picks up.

START—UP

The Valutrol DC SCR drive normally can be started up and made to run satisfactorily if the following step by step procedures are followed:

1. Verify that the terminal board screws and power connections are tight.

2. Verify that incoming power is the proper voltage and that the incoming wiring is complete and correct. Verify that the incoming reference voltages are correct.

3. With the diagnostic switch in its NORMAL position, close the circuit breaker and then apply power to the drive. If the green "ready-to-run" light located on the lower left hand corner of the main control card is not illuminated, the most probable cause is incorrect incoming phase rotation. Remove power, reverse any two incoming leads and repeat.

4. Turn the local speed reference **(LOC REF)** potentiometer on the diagnostic card to the center of its rotation and switch into the diagnostic run (DIAG RUN) position. As the local speed reference **(LOC REF)** potentiometer is rotated away from the control applying minus reference, the motor will begin to rotate. Check the tachometer polarity. With a DC tachometer, TKP is positive for rotation due to a minus reference at SR . Switch back to the normal position. If motor rotation was incorrect, remove power and interchange the fields F1 and F2.

5. Switch the diagnostic switch to the normal position. Run the drive from the normal reference up to top speed. Adjust **MAX SPEED**, if required.

6. Close and secure the front door of the panel unit.

SEQUENCE OF OPERATION

POWER APPLIED

The control transformer (CPT) is energized through its primary fuses. The blower will come on.

The power supply card is energized and the DC output (±20 volt) are applied through their fuses to the rest of the cards. All readings carry a tolerance of ±10%.

The motor field supply is energized. Refer to the motor field supply instructions for details.

If not faults have been detected by the monitor section of the Interface card, the fault relay FLT will close, FLT LED4 will illuminate and the "ready-to-run" indicator on the main control card will illuminate. Table IV tabulates the fault conditions which are monitored.

The oscillator will start, and the synchronizing signals SA, SB and SC will measure 8.5 volts RMS, (±10%). See Figures 15 and 17.

START (CONNECT RUN [OR JOG] TO —30 VOLTS)

SWITCH LOGIC

RUN or JOG will be switched from +30 Volts to —30 Volts (under purchaser control).

The MD control line MAC from the Main Control card to the Interface card will be pulled down to —20 Volts.

The Interface card checks that no faults exist and that relay C ON is picked-up before applying power to the coil of the M pilot relay MAX.

MAX picks up and applies power to the coil of the M relay. The M relay in turn applies power to the coil of the delay firing power relay (DFPR) which switches the DFP signal from the Main Control card to the Gate Pulse Transformer circuits on the Auxiliary Control card. Relay M also serves as a pilot relay for the DC contactor MD.

MD closes the motor armature loop and auxiliary contacts switch power to the coil of preconditioning relay PR. PR when picked-up releases the preconditioning signal PRE from common.

When PRE is released from common, it switches to —4 volts which will release the Main Control card preconditioning after approximately 80 milliseconds.

Releasing preconditioning allows the drive to send firing pulses to the gates of the SCR's in the conversion module, and allows the normal signal flow to occur.

SIGNAL FLOW

If RUN is switched, the reference at SR is applied to the linear time section. The timed reference output TR will ramp to a voltage proportional to SR. The REF SCALE adjustment is used to set TR to 10.0 volts when the input at SR is set for top speed. The time for TR to ramp from 0 to 10 volts is adjustable from .3 to 60 seconds with the LIN TIME adjustment. See jumper on system elementary (Ranges: .3 to 10 sec. or 2 to 60 sec.).

TABLE IV FAULT CONDITIONS

A fault has occured if the fault light (LED 4 is off) or if the READY TO RUN light is off. The conditions that can initiate a fault are as follows:

1. No three phase power
2. Loss of an incoming phase
3. Incorrect phase rotation
4. AC power fuse blown (if used)
5. Control fuse is open
6. Power supply plus or minus DC fuse is open

** 7. Instantaneous overcurrent (IOC) level exceeded.

8. Motor thermo-switch (OLD) (If connected to fault circuitry).
9. Timed overcurrent (TOC) - electronic. (If switching on the RBC card is closed).

**10. Loss of motor field

11. External Fault Stop momentarily released from Common.

*12. Other special functions to System Trip (SYSX) or External Fault Stop Inputs.

*13. System Trip input (SYSX) momentarily connected to +10 volts.

14. RESET button depressed or EXT. RSET input momentarily connected to 115V AC with motor rotating.
15. RESET button held depressed or EXT. RSET input held connected to 115V AC.
16. Diagnostic mode selected with the motor rotation.
17. Oscillator failed "on".
18. Tachometer fault (loss of tachometer signal)

**19. Overspeed

20. Trip actuated from ACC (by AC overcurrent or SCR commutation failure).

* May not be provided. Refer to the System elementary diagram.

** Can be used by **LOC REF** and **CUR REF** settings in Static Diagnostic mode.

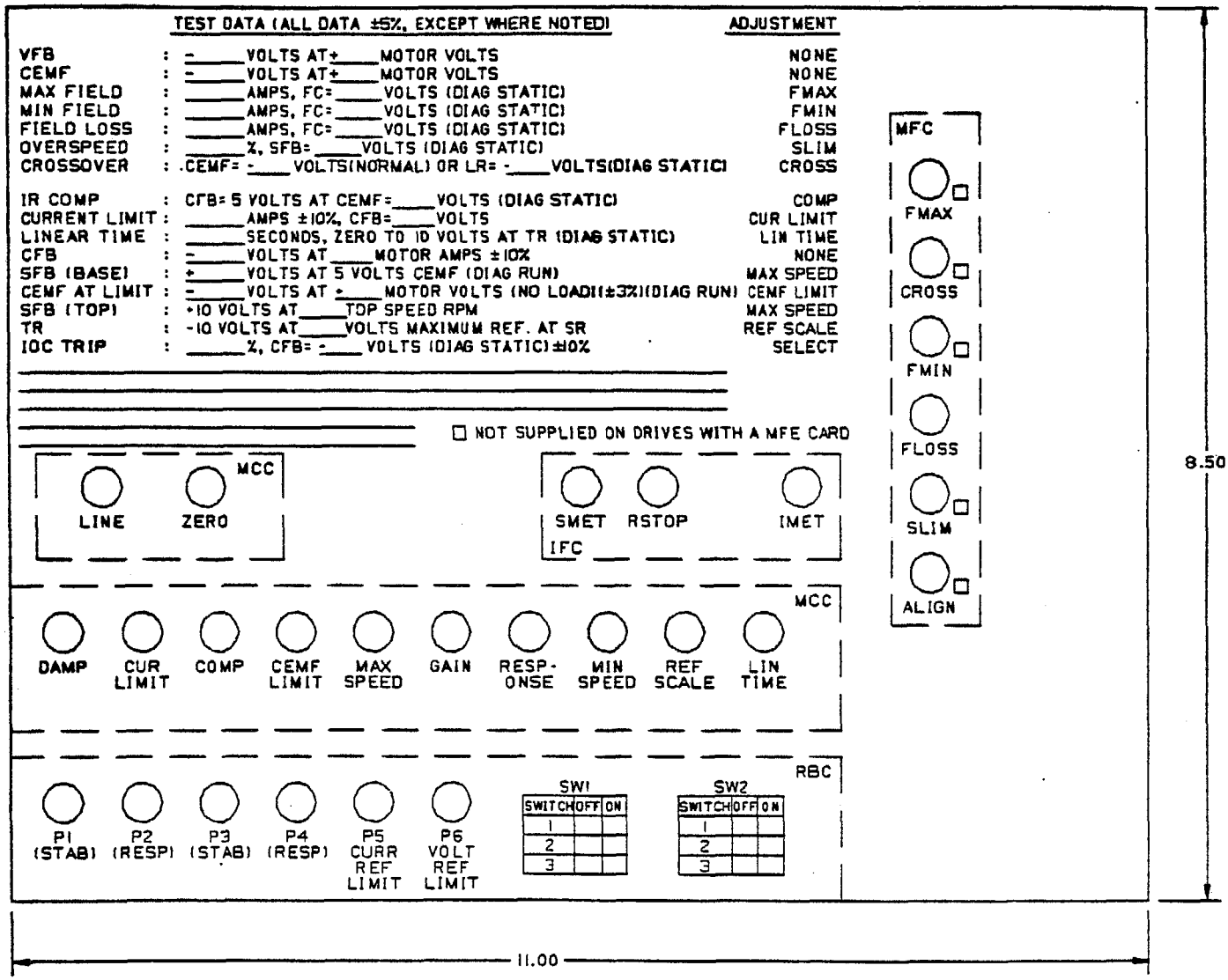
After the fault condition has been cleared and the motor has come to a standstill, the drive can be RESET by any of the following three methods.

1. Momentarily remove the three phase power and re-apply.

2. Push the RESET button

3. Momentarily connect EXT. RSET to 115V AC.

If all fault conditions have been cleared but the drive fails to RESET, the **RSTOP** adjustment may be set too low.

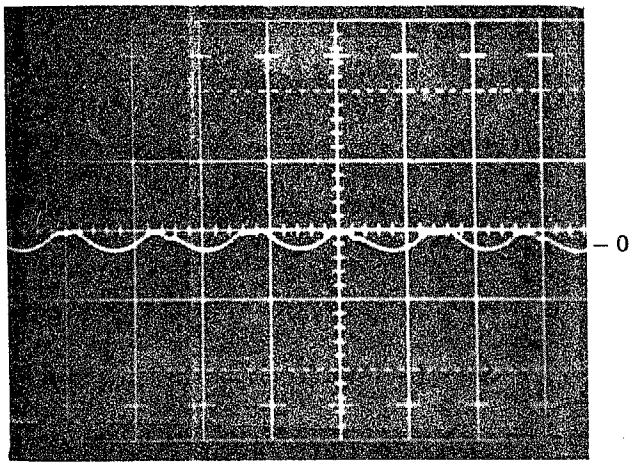


1. SIZE AS INDICATED.
2. FASSON WHITE CONVENTION BADGE (LABEL) STOCK 65P.
3. CUT TO SIZE.
4. SPRAY WITH PROTECTIVE COATING.

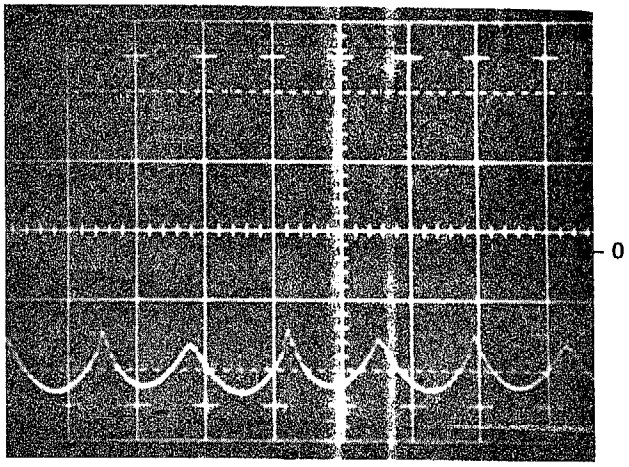
FIGURE 10
TEST DATA SHEET (36C766719AA)

WAVEFORMS

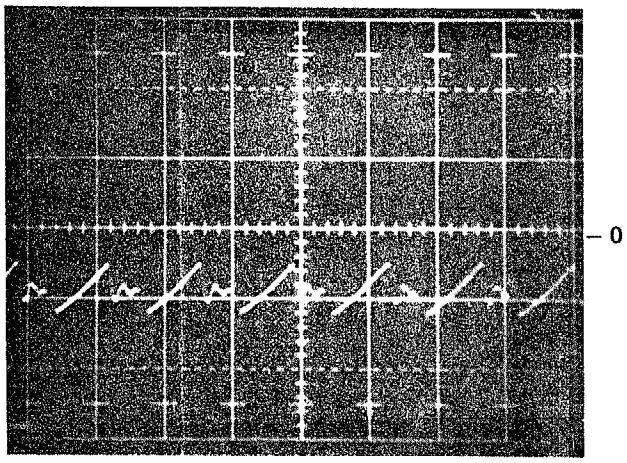
All illustrations were photographed in the forward motoring quadrant with zero volts on center line at 2 msec division.



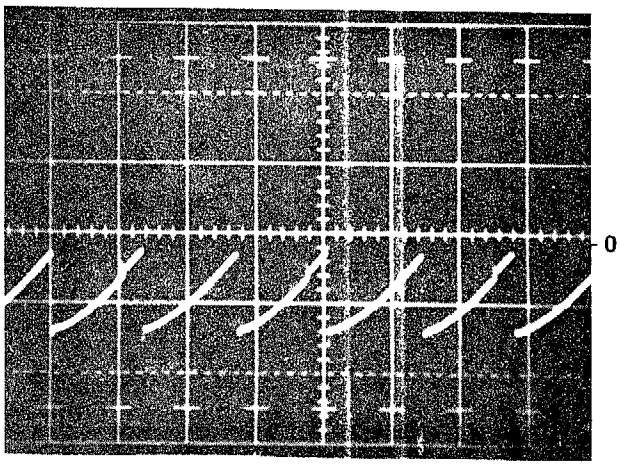
2 msec/div
At low current level 1 volt/division
Figure 11. Current Feedback (CFB)



2 msec/div
At continuous current 1 volt/division
Figure 12. Current Feedback (CFB)



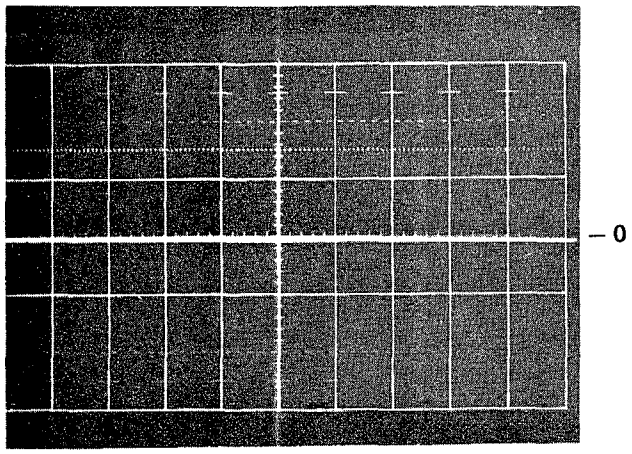
2 msec/div
At low current and 200 volts 5 volts/division
Figure 13. Voltage Feedback (VFB)



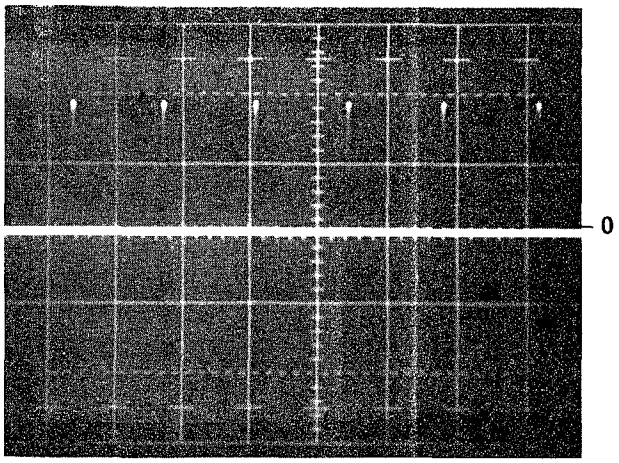
2 msec/div
At continuous current and 200 volts 5 volts/division
Figure 14. Voltage Feedback (VFB)

WAVEFORMS

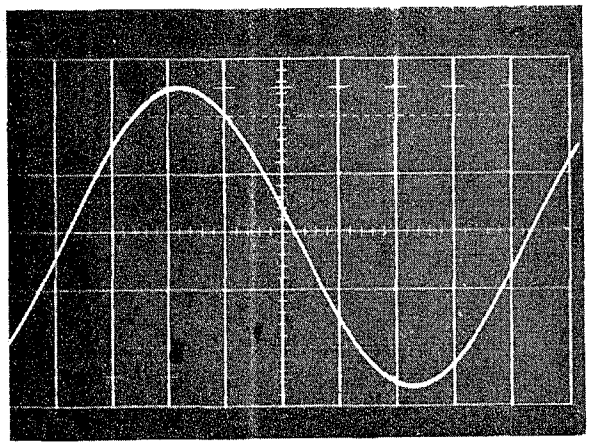
All illustrations were photographed in the forward motoring quadrant with zero volts on center line at 2 msec per division.



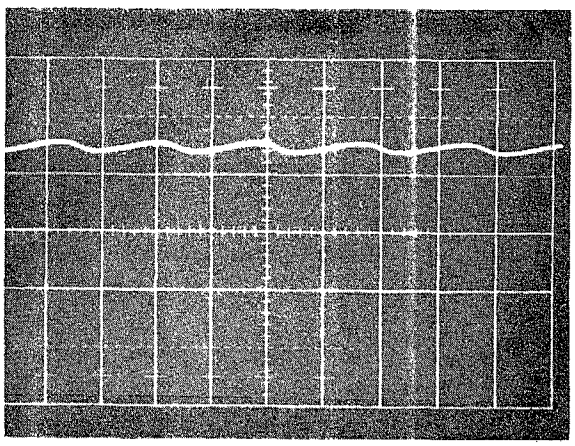
2 msec/div
(10 volts/division)
Figure 15. Oscillator (OSC)



2 msec/div
(2 volts/division)
Figure 16. Initial Pulse (IPU)



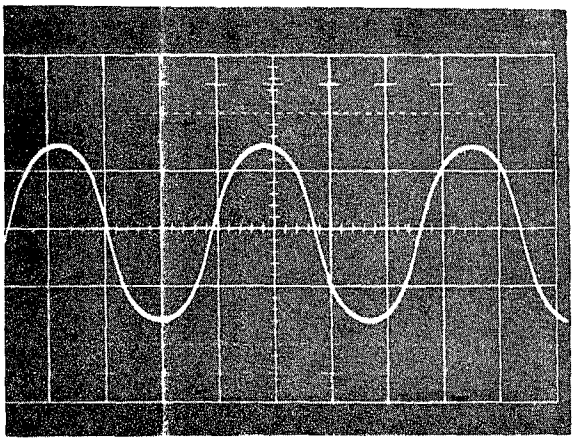
2 msec/div
Typical of SA, SB & SC
SB lags SA by 120°
SC lags SB by 120°
(5 volts/division)
Figure 17. Synchronizing Signal (SA)



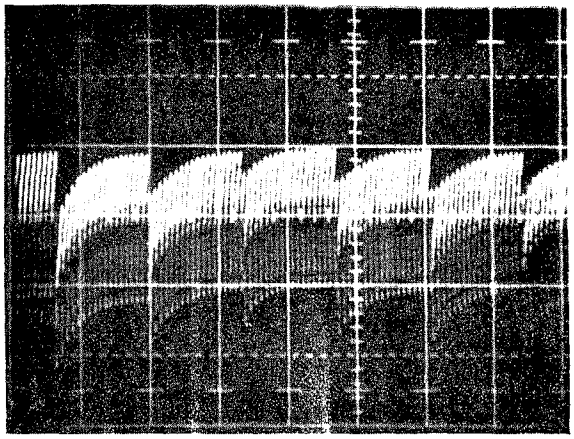
2 msec/div
With an AC tachometer at 450 RPM
(1 volt/division)
Figure 18. Speed Feedback (SFB)

WAVEFORMS

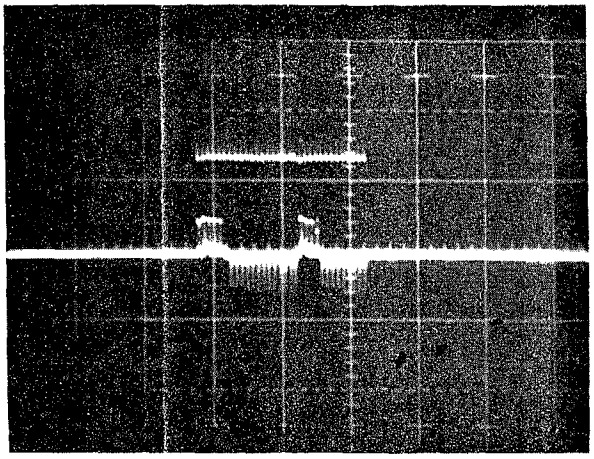
All illustrations were photographed in the forward motoring quadrant with zero volts on center line at 2 msec division.



2 msec/div
With an AC Tachometer at 450 RPM
(1 volt/division)
Figure 19. Tachometer Feedback (TFB)



2 msec/div
AC Coupled
(0.1 volt/division)
Figure 22. Pulse output (PO)



2 msec/div
Gate to Cathode Firing Signal 1G-1C
1 volt/division
Figure 23. Gate to Cathode Firing Signal

The speed (or CEMF feedback from the motor tachometer is scaled with a selectable resistor network on the interface card, and rectified (if required) on the main control card. Speed feedback may also be obtained by a jumper connection of the CEMF feedback signal into this input on the Interface card. The output of the speed feedback section is SFB, and will be 10 volts at top speed. **MAX SPEED** is adjusted to make the actual top speed correspond to desired top speed. See Figure 18.

The timed reference TR, and JOG reference JOGR, and the speed feedback SFB are summed by the regulator error amplifier. The error amplifier output EAO will be a low voltage (nearly zero) when the drive is regulating speed. EAO will not be low when the drive is in current limit or CEMF limit. The gain of the error amplifier is set with the **GAIN** adjustment on the MCC. The **GAIN** adjustment is used primarily to improve the response of the drive in the constant horsepower region when the motor field supply is a motor field control MFC.

To maintain good load regulation, the error amplifier is fed into the regulator integrator. The output of the integrator is the reference DR, to the driver. The response of the control below base speed is set with the **RESPONSE** adjustment on the MCC.

There is a limit however, to how responsive a drive may be set. Stability of the drive is generally decreased as its response is increased. The **RESPONSE** adjustment is desensitized when the drive is operating in the constant horsepower region of the torque speed curve.

To protect the system, three limit sections are provided; counter—EMF (CEMF) limit and two current limits. The output of the CEMF limit and the primary current limit drive the regulator integrator and will override the error amplifier, if required. The primary current limit is set with the **CUR LIMIT** adjustment and the counter EMF limit with the **CEMF LIMIT** adjustment on the MCC. Typcially the primary or regulator current limit is set at 150% of the motor nameplate current of 3.75 volts (±10%) of current feedback, CFB. For drives rated 500 volts DC the CEMF limit is normally 510 volts at no load or 5.60 volts (±10%) at CEMF. The secondary or driver current limit is in the driver section and will be described below. The primary current limit, if used, should not be set higher than the driver current limit. The driver current limit may be inhibited or adjusted as described below.

The main purpose of the secondary, or driver, current limit is to be an operating or fast response back-up current limit if the regulator, or primary current limit, is not used or allows current to overshoot its limit and the system is controlled by other signals summed at the driver junction.

The driver current limit may be increased by connecting a jumper from DCX to DCY on the main control card which will increase the limit approximately 130% to 175% of the motor nameplate current. Alternative levels may be established by connecting a resistor between DCL and DCX on the main control card. For normal levels of regulating current limit, either DCX should be jumpered on the DCY to increase the driver current limit or it should be inhibited by connecting DCI to COM. (See elementary diagram for specific application.) For regulating current limit above 150%, a resistor should be connected between DCL and DCX to raise the driver limit above 175% or it should be inhibited with a jumper between DCI and COM.

The counter—EMF signal CEMF is developed on the main control card by subtracting a signal proportional to IR drop of the motor from voltage feedback. This is set with the COMP adjustment.

The driver reference, DR, the voltage feedback, VFB, an armature current signal from damping adjustment DAMP, and the driver current limit output are summed at the input driver. The driver converts this error to pulse trains which drive the SCR gates in such a manner as to maintain the motor voltage proportional to the driver reference. The damping adjustment **DAMP** controls the response of the driver.

Generally speaking, **DAMP** is used to quiet small oscillations which occur in the current under light load conditions. Too much damping will slow down the system response and tend to cause overshoot.

The driver provides a signal IPU to the oscillator on the interface card to generate an initial pulse at the exact point in time that an SCR is to be fired. See Figure 16.

Two driver monitor points are available, PCR and PO. PCR is the phase control reference which causes the output pulse trains to phase shift in time with respect to the AC line. As PCR moves from zero to +6 volts (±10%), the output pulses will shift from full off to full on. PO is used to monitor the pulse outputs to the SCR's. See Figure 22.

STOP

There are two stop sequences, normal stop and fault stop. With a normal stop the drive regenerates to near zero speed before opening the MD contactor. A fault stop opens the contactor and drive coasts to a stop.

Normal stop (disconnect RUN from —30 volts).

RUN will switch from —30 volts to +30 volts. MAC will switch to zero volts and the system reference input to the linear time section will be shunted to common.

The timed reference TR will begin to time down to zero and the drive speed will come down accordingly.

The regenerative stop circuit on the interface card will hold the contactor closed until the CEMF signal is almost zero, corresponding to zero speed. At this time, the preconditioning signal PRE goes to common, removing power from the MAX coil. 100 milliseconds later, MAX drops out removing power from M, which then drops out and in turn drops out the MD contactor. The CEMF level corresponding to zero speed is set by the **RSTOP** adjustment. If **RSTOP** is set too far (CW) power is removed prematurely and the drive will coast into zero speed. If **RSTOP** is set too far (CCW) the MD contactor may not open at all.

In some cases the regenerative stop circuit, (described above) may be under the control of the speed feedback signal, SFB, rather than the CEMF signal.

FAULT STOP — Fault detected (See Table IV) or CONTROL ON. (C ON) is dropped out.

The preconditioning signal PRE is immediately applied to the main control card, forcing the drive into zero current or coast conditions. As soon as the current goes to zero, preconditioning is established throughout the card.

The M relay unconditionally drops out 100 milliseconds after the fault condition occurs and in turn drops out the MD contactor.

The drive can not be restarted until the motor has come to rest. If the STOP was initiated by a fault, this is taken care of automatically, but it is the purchaser's responsibility to **not** pick-up "C ON" before the motor has come to rest. After the motor has stopped, push the RESET button.

DIAGNOSTIC STATIC (SWITCH TO LEFT)

LOGIC

The RUN and JOG inputs are inhibited. This prevents the references SR and JOGR from activating the drive and holds the MD contactor open.

The current reference potentiometer **CUR REF** controls the current feedback signal CFB.

The local reference **LOC REF** potentiometer is connected into the input of the linear time section and into the speed feedback section. The local reference is also connected to the field diagnostic reference FDR. Refer to motor field control instructions (GEK-24971) for details of operation.

To simplify signal tracing, the gain of the regulator and drive is reduced and the speed feedback signal to the regulator error amplifier is removed.

SIGNAL FLOW

The local reference LR is applied directly to the input of the linear time section, by-passing the **REF SCALE** adjustment. The timed output TR will ramp to a voltage equal to LR in magnitude and polarity with time determined by the setting **LIN TIME.**

The local reference LR is also applied to the input of the last stage of the speed feedback section. The output SFB will be equal to LR in magnitude, but opposite in polarity. The tachometer scaling circuit and its output TFB are unaffected by the local reference and will remain at zero. As the signal from SFB into the regulator error amplifier is inhibited the primary purpose of exercising SFB is to check those external special function circuits which are programmed from SFB, and/or SFB functions of an MFC.

A dummy feedback signal to replace the normal SFB signal is connected from the regulator integrator output DR to the input of the regulator error amplifier. Under these conditions DR is equal to the magnitude of TR but opposite in polarity as long as the current reference is below the current limit setting. As the current reference is raised, the current feedback signal CFB will exceed the current limit level set by **CUR LIM** and force the DR output into negative saturation for forward current limit and positive saturation for reverse current limit.

Current feedback will also program the CEMF output to a level proportional to the CFB level and the **COMP** adjustment.

The load instrument output IMET will also respond to the current reference.

The gain of the drive is reduced so that the phase control reference PCR is equal to the magnitude of the driver reference DR as long as the current reference is set at zero.

With an oscilloscope, the initial pulse output IPU, may be monitored to verify proper operation. See Figure 16.

DIAGNOSTIC RUN (SWITCH RIGHT)

(Diagnostic mode may be inhibited by lack of jumper on the Diagnostic card. Refer to system elementary and operating notes.) In diagnostic run, the local reference LR and the diagnostic switch are substituted for the reference(s) SR, JOGR and the RUN and JOG switch inputs just as in diagnostic static. The drive then runs normally with one important exception: system feedback is normal but the signal from system feedback to the regulator error amplifier is inhibited and the dummy feedback from DR is still in place.

The net effect is the drive operates as a base speed voltage regulated from the **LOC REF** potentiometer.

CALIBRATION PROCEDURE

The diagnostic card is used to generate the appropriate test signals and operating modes to calibrate the drive.
To avoid confusion and possible interaction, the adjustments should be made in the following sequence.

CAUTION

DO NOT CHANGE SWITCH SETTINGS WHILE THE DRIVE IS OPERATING.

TABLE II
Recalibrating Adjustment Sequences

	WITH MOTOR FIELD CONTROL
DIAGNOSTIC STATIC MODE, ADJUST	**LINE** **FMAX** **FMIN*** **FLOSS** **SLIM** **CROSS*** **LIN TIME** **COMP** **CUR LIMIT (MCC)** **IMET** (IF USED)
DIAGNOSTIC RUN MODE, ADJUST	**MAX SPEED** **ALIGN** **CEMF LIMIT** **SMET** (IF USED)
NORMAL MODE, ADJUST	**REF SCALE** **MAX SPEED** (TRIM) **MIN SPEED** (IF USED) **GAIN (MCC)** **RESPONSE (MCC)** **DAMP** **RSTOP**

All of the high voltage inputs to the controller have been scaled down with the scale factors shown on the test data sheet.

CALIBRATION WITH EXTERNAL MOTOR FIELD CONTROL (MFC)

All readings can have a tolerance of ±10%.

Select Diagnostic static and set the **CUR REF** and **LOC REF** to the center positions.

LINE (Line Impedance Compensation)

This function is factory set at approximately mid-range (3 1/2% impedance). CW rotation of **LINE** potentiometer adjusts for greater line impedance. Range is 2% to 5% with no jumper RLA to PCR. Range is 4% to 10% with jumper RLA to PCR. If impedance is unknown a mid-range setting, no jumper is suggested.

FMAX (maximum field)

Set the **LOC REF** potentiometer for —1 volt at LR. Adjust **FMAX** until FC corresponds to the maximum field FC on the test data sheet.

FMIN (minimum field — limit)

Set **LOC REF** potentiometer for —7 volts at LR. Adjust **FMIN** until FC corresponds to minimum FC on the test data sheet.

FLOSS (field loss — fault)

Set the **LOC REF** to center position and reset the drive. Adjust **FLOSS** full CW.

Monitor FC and move the **LOC REF** potentiometer Rev until FC corresponds to the field loss value on the test data sheet. Slowly rotate **FLOSS** CW until the "Ready to Run" light turns off indicating a drive fault. Reset the drive.

SLIM (speed limit — overspeed fault)

Set the **LOC REF** to center position and reset the drive. Adjust **SLIM** full CW.

Monitor SFB and move the **LOC REF** potentiometer forward until SFB corresponds to the overspeed limit on the test data sheet. Slowly adjust **SLIM** CCW until the "Ready to Run" light turns off indicating a drive fault.

CROSS (cross over — field)

Set **CROSS** full CCW. Turn the **LOC REF** potentiometer forward until LR corresponds to the cross over LR on the test data sheet.

Monitor FC and adjust **CROSS** CW until FC just starts to increase. **CROSS** may be checked when the drive is running in normal operation by verifying that CEMF reads the value on the test data sheet with the drive operating above base speed.

LIN TIME (linear time)
Monitor TR and set to zero with the **LOC REF** potentiometer. Rapidly turn the **LOC REF** full Fwd and measure the time for TR to ramp to 10 volts (±10%). Adjust **LIN TIME** until this time corresponds to the test data sheet linear time.

COMP (compensation — IR)
Set the **LOC REF** potentiometer to center position. Adjust the **CUR REF** potentiometer Fwd or Rev until CFB is at 5 volts (±10%).

Monitor CEMF and adjust **COMP** until **CEMF** equals the value on the test data sheet.

CUR LIMIT (current limit)
Set **CUR LIMIT** fully CW. Adjust the **CUR REF** potentiometer until CFB corresponds to the current limit level on the test data sheet. Monitor DR and turn **CUR LIMIT** CCW until DR just moves away from zero.

IMET (load instrument calibration)
Adjust the **CUR REF** until CFB corresponds to full load current. Verify that the optional load instrument reads full load. If not, remove power; adjust **IMET** and repeat.

Set the **LOC REF** to the center position; reset the drive and switch to Diagnostic Run.

CEMF LIMIT (counter emf limit)
Turn **CEMF LIMIT** full CCW and turn the **LOC REF** potentiometer full Fwd. Adjust **CEMF LIMIT** until CEMF corresponds to the CEMF LIMIT on the test data sheet.

MAX. SPEED/ALIGN (max speed/tach loss alignment fault)
Turn **MAX SPEED** full CCW. Turn **ALIGN** full CW. Adjust the **LOC REF** potentiometer until CEMF reads 5 volts (±10%). Adjust **MAX SPEED** until SFB corresponds to the base speed feedback on the test data sheet.

Monitor TA and adjust **ALIGN** CCW until TA is approximately zero volts.

SMET (Speed instrument calibration)
Turn the **LOC REF** potentiometer until SFB is 3 volts (±10%). The optional speed indicator should indicate 30% top speed. If it does not, push **RESET** button to initiate a shut down. Remove power, adjust **SMET** and repeat.

Return the Diagnostic switch to Normal.

REF SCAL/MAX SPEED (reference scale/max speed)
Turn **REF SCALE** full CCW. Start the drive and apply top speed reference to SR. Adjust the **REF SCALE** potentiometer until SFB is 10 volts (±10%). This normalizes the timed reference TR and speed feedback, SFB for 10 volts (±10%) at top speed.

Now measure motor RPM and adjust **MAX SPEED** (if necessary) until the actual RPM corresponds to the desired top speed. If actual top RPM was off by more than 5% reset **ALIGN** as detailed above.

RSTOP (regenerative stop)
With the motor operating at some RPM, call for a drive stop by initiating the proper magnetics which will release 2TB-14 from —30 volts. The motor will decelerate to a low speed and the MD contactor will open. If the MD contactor opens before the drive comes down to a stop, **RSTOP** is set too high. If MD contactor fails to open, **RSTOP** is set too low. Push the RESET button to drop out MD prior to removing power. **RSTOP** should be readjusted with power removed. Turn **RSTOP** CW to drop out MD at a higher speed.

MIN SPEED (minimum speed)
Reduce the system reference to minimum and start the drive. Adjust **MIN SPEED**, as required, to meet system miminum requirements. Refer to system elementary for circuit details.

GAIN, RESPONSE, DAMP and COMP (Stability adjustments on the MCC card)

1. Set **DAMP** potentiometer at minimum 7 o'clock position.

2. Place the Diagnostic switch in the static mode Adjust **CUR REF** for 2.5 volts at test point CFB. This is equivalent to rated armature current.

3. For DC motors in the range 400 thru 900 HP, 500V DC thru 630V DC, set **COMP** potentiometer by reading the voltage CEMF pin and obtaining a value as follows:

MOTOR BASE SPEED	CEMF VOLTAGE
Below 400 RPM	0.444
Above 400 RPM	0.296

The **COMP** potentiometer is now set for proper operation. Regardless of overload range, IOC setting or motor field range this setting is correct and should not be changed.

4. Set the **GAIN** adjustment by calculating the GAIN number and referring to the chart. (Figure 24)

$$\text{Gain No} = \frac{\text{Maximum Operating Speed}}{\text{Motor Base Speed}}$$

See motor nameplate under — Speed
Motor Base Speed/Maximum Operation Speed
Example: 1150/3600 RPM

5. Set **LIN TIME** potentiometer at minimum (7 o'clock)

6. Set **RESPONSE** potentiometer at minimum (7 o'clock)

When drive is functioning properly in all other respects make small incremental step increases and decreases in speed below base speed. Observe armature current for bumping repeatedly before steady state speed is attained.

Increase the **RESPONSE** setting (move CW) until bumpy setting current is observed. Then reduce the **RESPONSE** setting until no bumps (or only one) is observed. This is the maximum **RESPONSE** setting.

In general, settings below 10 o'clock will show signs of increasing sluggishness. Settings greater than 2 o'clock may show signs of hard or even continuous bumping. Full **RESPONSE** setting (5 o'clock) will usually trip the IOC.

7. RESET **LIN TIME** to required setting.

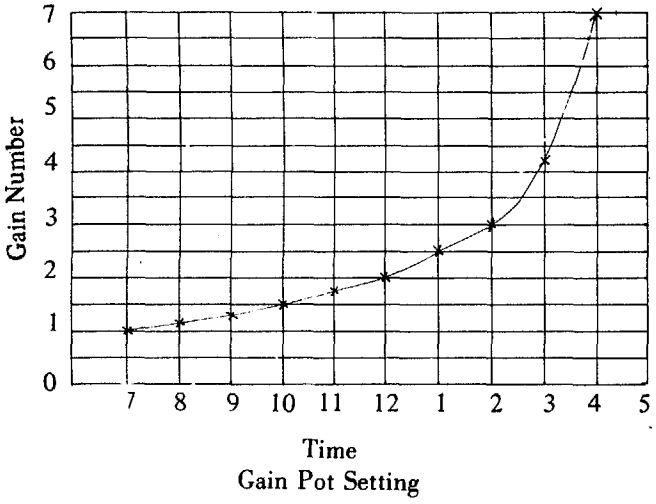


Figure 28
Gain Adjustment

After this setting has been made, make sure no further adjustments to the Gain Pot.

For drives utilizing the inner current loop circuitry provided on the Regulator/Buffer card, the following adjustments on the Main Control card will be inoperative:

GAIN
RESPONSE
CUR LIMIT
CEMF LIMIT

If the Inner Current Loop is used, the following switches will be closed on the RBC card:

SW1-1
SW1-2
SW1-3

Refer to the test data sheet for switch position. The following jumpers will also be in place if the Inner Current Loop is used.

MCC — RJR to AK — RJR2

MCC — RERR to MCC — COMM

MCC — DRX to AK — DRX2

RBC — CRA to RBC — CRB
(may be a resistor R75, in place of jumper)

RBC — VRA to RBC — VRB
(may be a resistor R76, in place of jumper)

Potentiometers 1 thru 4 on the Regulator/Buffer card are used to set the stability and the response of the regulator. Potentiometer **P5 (CURR LIMIT)** is used to adjust current limit and potentiometer **P6 (VOLT LIMIT)** is used to adjust, voltage limit. The factory settings for these potentiometers and switch positions are indicated on the test data sheet.

CAUTION

DO NOT CHANGE SWITCH SETTINGS WHILE THE DRIVE IS OPERATING.

TROUBLESHOOTING

Although many of the problems which may arise can be effectively located with a multi-meter, an oscilloscope is a very powerful troubleshooting tool. The only requirements are that the selected scope have a DC input capability and a line synchronization mode. Caution should be exercised in measuring any point with a possible high potential with any instrument; however, particular care should be taken with an oscilloscope since the common clip is normally connected directly to the instrument case. If the grounded plug has not been defeated it will cause a short circuit between the high potential point under test and ground.

RECOMMENDED INSTRUMENTATION

Simpson Multi-meter (or equivalent). 10,000 ohms/volt (or higher).

Hewlett-Packard or Tektronix (or equivalent). Dual Trace oscilloscope rate for operation from DC to 10 MHZ at 0.01V/CM with deflection factors to provide 0.01 V/cm to 1300V peak to peak deflection when used with appropriate attentuator probes.

PROCEDURES

In troubleshooting this drive system the most appropriate place to start is to follow the SEQUENCE OF OPERATION (previously described) until a discrepancy or fault is noted. This step by step procedure will determine which part, sub assembly or printed circuit card is causing the problem.

Included in this procedure is the use of the built-in Diagnostic card (DGC). This is another powerful tool for quickly locating drive system faults.

If the malfunction is a performance problem, then the quickest way to discover the problem is to follow the CALIBRATION PROCEDURE (previously described).

Detailed adjustment for the MFC cards are found in GEK-24971.

HOW TO TEST AN SCR

WARNING

ELECTRIC SHOCK CAN CAUSE PERSONAL INJURY OR LOSS OF LIFE. WHETHER THE AC SUPPLY IS GROUNDED OR NOT, HIGH VOLTAGES TO GROUND WILL BE PRESENT AT MANY POINTS THROUGHOUT THE SYSTEM.

1. Disconnect the AC power and make sure the loop contactor (MD) is open.
2. Using a multi-meter selected to read ohms on the times 1K scale, check the forward and reverse resistance of SCR cell pairs. This is done first by reading across power terminals L1 and P, L2 and P, L3 and P. Then read across power terminals L1 and N, L2 and N and L3 and N. Be sure that the LEG fuses (FU1—FU6) are not open in the above.

SCR Description	Forward Reading	Reverse Reading
Good SCR	100K to Infinity	100 K to Infinity
Shorted SCR	Zero	Zero
Inoperative SCR	1 to 2K	100K to Infinity
Open SCR	100K to Infinity	100K to Infinity

3. Since an open SCR will give about the same resistance reading as a good SCR another method must be used to find this type of fault. It should be pointed out, however, that practically all cells fail by shorting and very few by opening. If an open SCR is suspected or it is desired to check the switching operation of an SCR, the following circuit should be used:

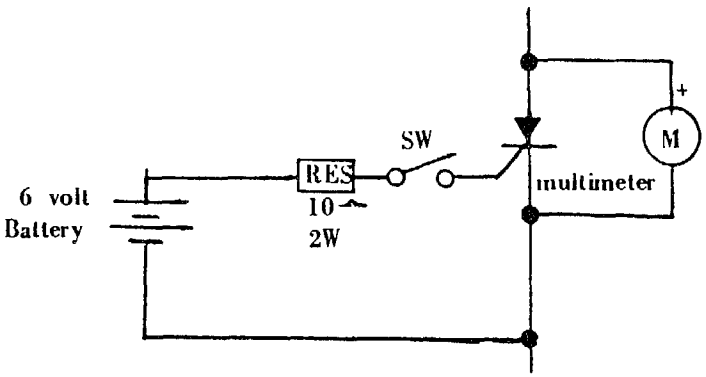


FIGURE 25

The multimeter is selected to read ohms on the 1K scale, and is connected to read the forward resistance of the SCR. When switch SW is closed, the forward resistance of a good SCR will change from a high value (100K to infinity) to a low value (1 to 10K). When the switch is opened a good SCR will revert to its high forward resistance or blocking state, if the holding current (multi-meter battery) source is momentarily removed. A faulty SCR will not switch remaining in either an open or a conducting state.

4. If any SCR s are suspected of being faulty from the above resistance checks, the appropriate SCR stack should be removed from the case. After the SCR cathode and gate plugs have been disconnected, recheck the forward and reverse resistance before replacing the SCR heatsink assembly. This should be done before any SCR is definitely classified as damaged or faulty, since a fault in another SCR or another part of the circuity can produce faulty reading from a good SCR before it is disconnected from the circuit.

REMOVAL/REPAIR

SCR STACK REMOVAL PROCEDURE

Remove the glastic baffels covering the stacks(s) to be removed, see Figure 26. Next, remove the leg fuses or bus straps connected on the upper and lower sections of the stack assembly. Disconnect the SCR gate and cathode plug on each end of the stack. Disconnect the SCR stack assembly from the bridge and place it on a clean work table, see Figure 27. Note and record exactly where all hardware and wires go. If the stack assembly is dirty, clean it using a soft paint brush, cleaning cloths, and low pressure dry air or Freon solvent spray.

Check and record the reading of the spring bar gauge on the side of the stack assembly containing the defective cell, see Figure 30. During reassembly, you must adjust the heatsink clamp pressure so that the spring bar gauge reads this same value. Be sure to sight across both ends of the spring bar or the reading will be inaccurate.

Place the stack assembly on its side and disassemble the portion of the assembly containing the defective SCR cell. Alternately loosen each nut in the clamp assembly to keep even pressure on the SCR cell. As the various parts are removed from the stack, place those parts on a clean surface in the same orientation and order in which they were removed. See Figure 28.

Remove the defective cell. Clean and lubricate the heatsinks before reassembling. Place a few drops of GE SF1153 silicone oil in the center of both heatsink mounting surfaces. Polish the interface area until clean using a combination of back and forth, sideways, and circular motions with a lint-free cloth or tissue. Use a clean, lint-free cloth or tissue to wipe the surfaces clean. Finally place a few drops of GE SF1153 silicone oil on the mating surfaces of the heatsink and use a lint-free cloth or tissue to spread the oil evenly. Wipe off excess oil.

Clean and lubricate the SCR cell. Place the cell on a flat table and clean the anode and cathode faces with GE SF1153 oil as described above. Apply a thin film of silicone oil to cover the mating surfaces and wipe off any excess.

Coat the threads, nuts and washers used in the clamp mechanism with a small amount of graphite spray.

Replace the SCR cell carefully, ensuring that is orientation and polarity are correct. Use care when placing the cell on the dowel pin to prevent scratching the cell post surface. Rotate the cell back and forth a few times to ensure proper seating.

Reassemble the SCR cell assembly, taking care not to damage the cell with the heatsink. Tighten the clamp nuts only finger tight. Ensure that the number of exposed threads on each stud are about equal. Tighten each nut to 12 inch pounds to take up all the slack in the assembly. Tighten each nut alternately with approximately.one-half turns.

Until a total of 1-3/8 turns has been made, sight across both ends ot the spring bar and verify the guage is at its original setting.

Replace the SCR cell stack assembly in the bridge, being careful to properly reconnect all wires and replace all hardware.

NOTE

THERE SHOULD BE NO NEED TO RETUNE THE DRIVE AFTER REMOVAL/REPAIR OF A CONVERSION MODULE. AN SCR OR ANY OTHER REMOVABLE SUB—ASSEMBLY UNLESS OF COURSE AN ADJUSTMENT WAS INADVERTENTLY MOVED OR DISTURBED. IF A PRINTED CIRCUIT CARD IS REPLACED (OTHER THAN THE POWER SUPPLY CARD PSC):

1. ADD STAB—ON JUMPERS TO THE REPLACE—MENT CARD JUST LIKE THE JUMPERS ON THE CARD THAT WAS REPLACED OR AS LISTED ON THE SYSTEM ELEMENTARY DIAGRAM PROGRAMMING TABLE.

2. ADD STAB—ON RESISTORS AND CAPACITORS TO THE REPLACEMENT CARD JUST LIKE THE COMPONENTS ON THE CARD THAT WAS REPLACED OR AS SHOWN WITH VALUES ON THE SYSTEM ELEMENTARY MAIN CONTROL CARD (MCC) AT STAB—ON TERMINALS TL, RJ, SFB, NDE, CL1, LT AND CLJ OR ON THE DM1, DM2, ETC. TERMINALS OR ANY OTHER PRINTED CIRCUIT CARD.

3. SET THE POTENTIOMETERS ON THE REPLACEMENT PRINTED CIRCUIT CARD TO THE POSITION AS WAS SET ON THE CARD THAT WAS REPLACED OR TO THE POSITION SHOWN ON THE TEST DATA SHEET. RECHECK THE RECALIBRATION PROCEDURES DESCRIBED.

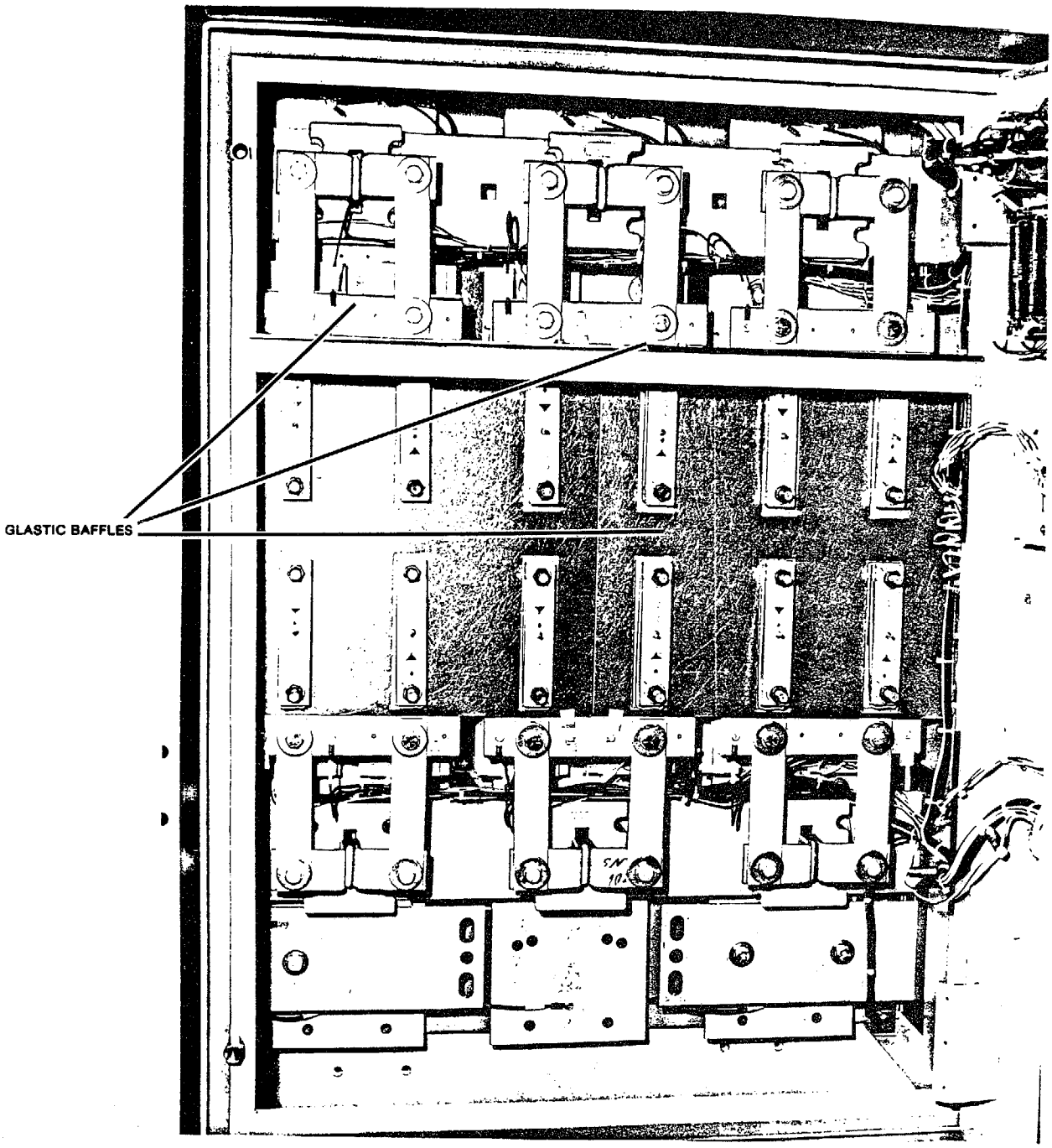


FIGURE 26
POWER UNIT (inside view)

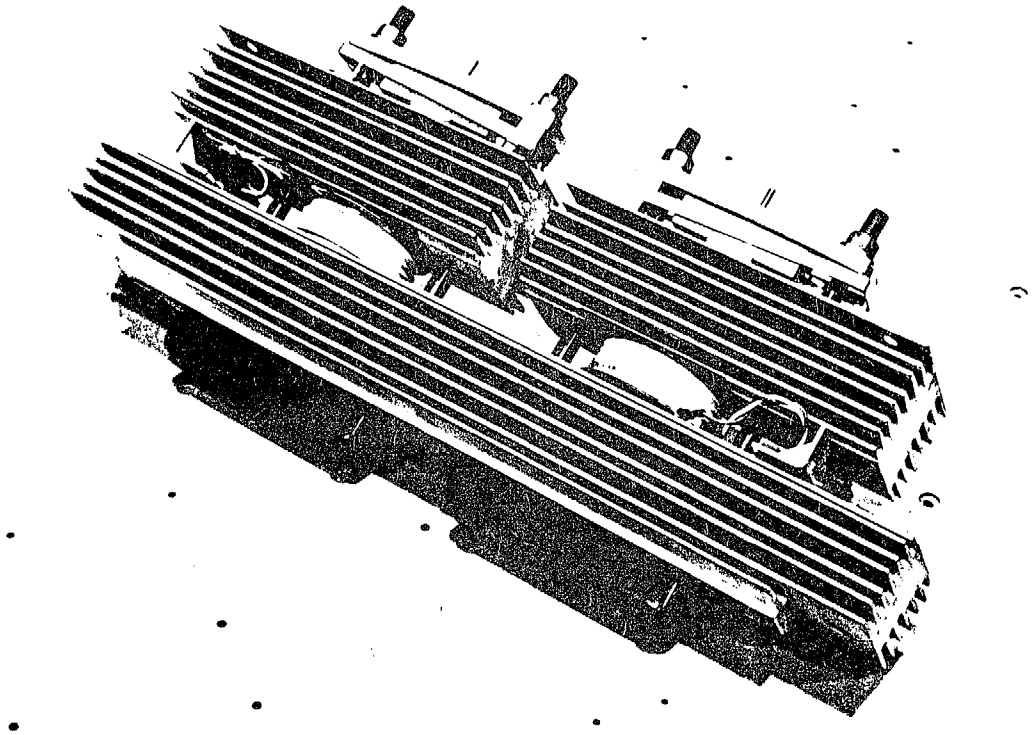


FIGURE 27
SCR CELL STACK ASSEMBLY

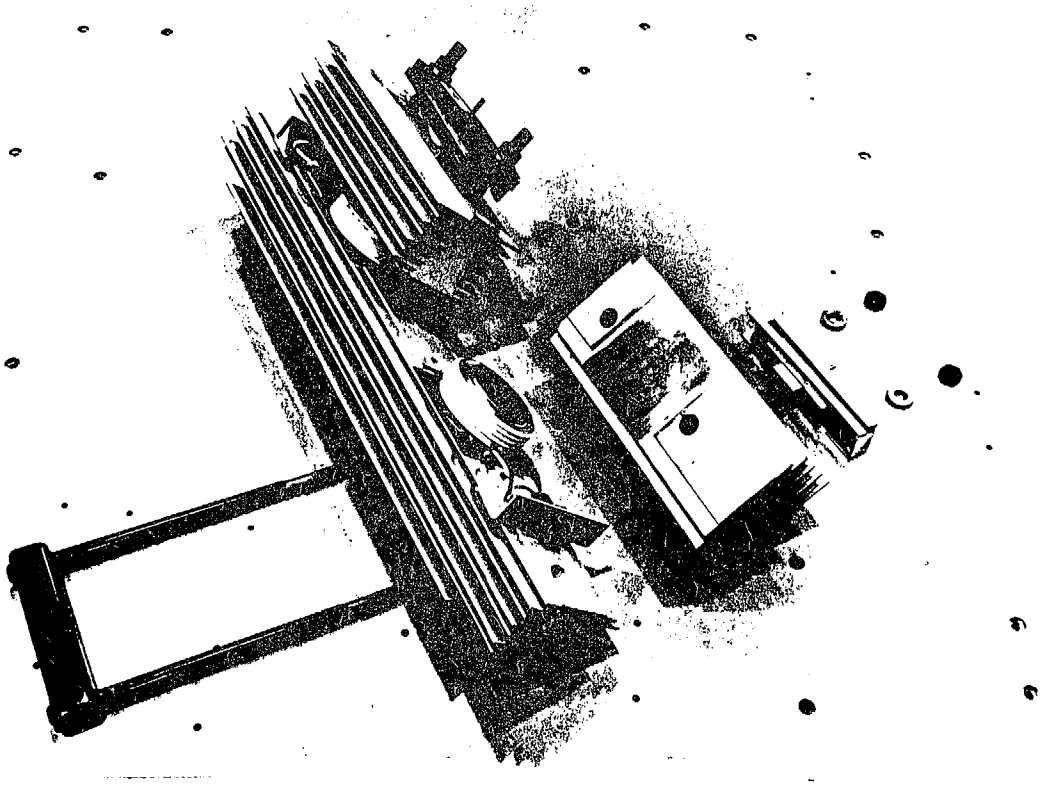


FIGURE 28
SCR STACK DISASSEMBLED

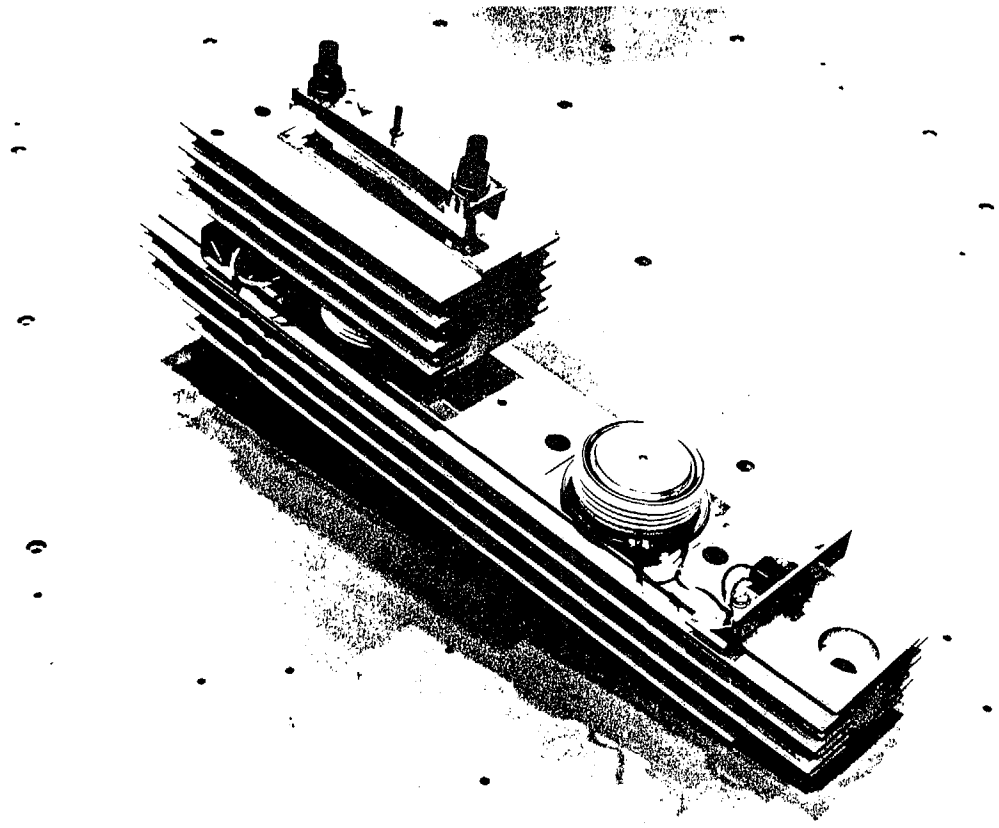


FIGURE 29
SCR STACK — LESS HEATSINK

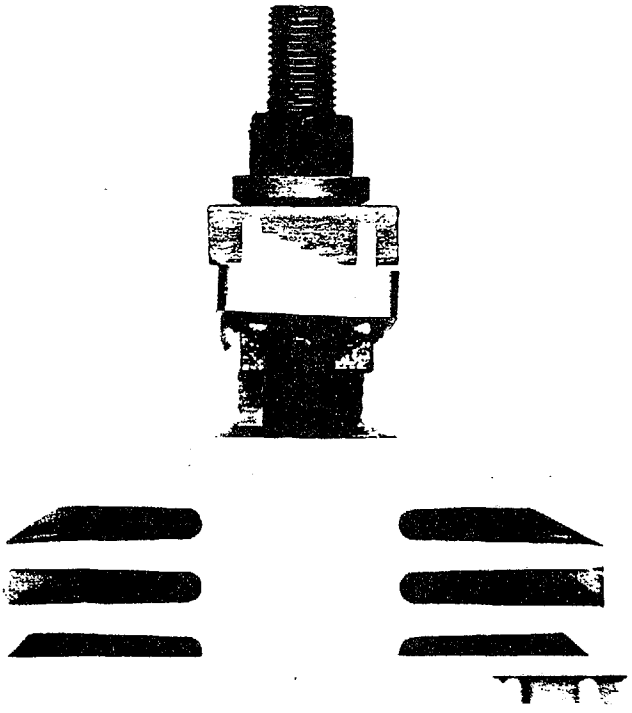


FIGURE 30
SPRING BAR GAUGE

GENERAL ELECTRIC COMPANY
SPEED VARIATOR PRODUCTS OPERATION
ERIE, PENNSYLVANIA 16531

GENERAL ELECTRIC

GEK-85760 (10/83) 200 (P)